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FINAL REPORT

AI Washing or Credible Disclosure? Patterns and Clusters in Corporate AI Disclosures

By

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I declare that this assignment is my own work, which all sources of reference are acknowledged in full, and it has not been submitted for any other course.

Signature:

How AI disclosure is measured

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1. INTRODUCTION

1.1 BUSINESS PROBLEM

The release of ChatGPT in November 2022 turned artificial intelligence from a specialist topic in technology-sector filings into a disclosure obligation for essentially the entire S&P 500. Between 2021 and 2025 the share of S&P 500 firms making at least one AI-related statement in their SEC filings rose from 36% to 92%, and the narrative density of AI disclosure (AI statements per 1,000 words of corporate narrative text) grew nearly seven-fold (from 0.016 to 0.111 per 1,000 words). The single largest jump occurs in the 2024 filing season, when fiscal-year-2023 10-Ks become the first annual reports written with a full year of generative AI behind them: the share of 10-Ks containing any AI statement jumps from 39% to 72% in one season (Eisfeldt, Schubert, and Zhang 2023; Babina et al. 2024).

This is, at the same time, an investor problem, an issuer problem, and a regulatory problem:

- **Investors** need to know whether a firm's AI narrative reflects real capability, real deployment, and real economic exposure, or whether it is costless positioning attached to a hot topic (Basnet et al. 2025; Song et al. 2026).
- **Issuers** need to know how much disclosure is enough, and in which venue, without inviting enforcement risk.
- **Regulators**, the SEC prominently among them, have already acted. Chair Gensler's December 2023 warning about "AI washing" (Gensler 2023) and the March 2024 enforcement actions against Delphia and Global Predictions (U.S. Securities and Exchange Commission 2024) made explicit that overstating AI capability in disclosure and marketing material is a securities-law violation, not merely a reputational risk.

The correspondence between the SEC and Welltower over its 2024 10-K is the clearest illustration of the underlying concern (Strauss et al. 2025): the regulator did not object to promotional language in the filing itself, but to earnings-call and press-release language ("industry-leading," "competitive advantage") that was not backed by proportional disclosure in the 10-K. The mechanism the SEC is policing is a gap between venues, not a gap between words and an unobservable truth.

Mention counts, the workhorse measure of the finance and disclosure literature, tracing back to Loughran and McDonald's (2011) word lists, cannot answer any of these questions with the precision the problem now requires. A count of how many times "artificial intelligence" appears in a 10-K conflates a risk-factor boilerplate sentence with a specific claim about a named, deployed product generating measured revenue. It cannot:

- separate promotional register from governance register;
- tell whether a claim describes something the firm has done versus something it might do; or
- compare what a firm says in a call against what it commits to in a filing.

Answering the business problem (is the disclosure substantive, and does it matter economically) requires moving past counting mentions and building a measurement pipeline that extracts what is

actually claimed: who is the subject, what type of AI, in what tense, in what register, and, one layer deeper, what concrete activity, with what evidence, is behind the claim.

1.2 RESEARCH QUESTIONS

The overarching question this thesis answers is: **how do public firms differ in how they disclose AI activity (adoption, capability, risk, and governance) and does the content of that disclosure carry economic information beyond how much of it there is?**

This decomposes into six subsidiary questions, each mapped to a distinct empirical design in the chapters that follow:

- **RQ1: Disclosure structure and archetypes.** How do firms differ in the volume and composition of AI disclosure, and do those differences organize into recurrent, stable archetypes, or is the population better described as a continuum?
- **RQ2: Voice vs. disclosed behaviour.** Within a firm's own disclosure, does promotional voice exceed the concrete behaviour the firm describes, and is that gap stable and interpretable, or is it noise?
- **RQ3: Economic coherence.** Do disclosure archetypes correspond to economically distinct types of firm (in R&D intensity, margins, risk, and valuation) even after controlling for sector and year, or is the segmentation merely a taxonomy of talk?
- **RQ4: Incremental information.** Beyond how much a firm says about AI and beyond its financial fundamentals, does the semantic content and the disclosed activity of what it says carry incremental information about systematic risk, valuation, R&D intensity, and future growth?
- **RQ5: Venue consistency.** Does the same firm, in the same fiscal year, disclose AI consistently across venues (annual report, quarterly filing, proxy statement, current report, and earnings call), or is there a structural gap between what is said to analysts and what is filed with the SEC?
- **RQ6: Evolution and scrutiny.** How has AI disclosure evolved from 2021 to 2025, and is that evolution attributable to regulatory scrutiny (the SEC's AI-washing enforcement of 2024), or is it inseparable from the broader adoption boom?

1.3 LITERATURE ON CORPORATE AI DISCLOSURE AND AI WASHING

1.3.1 THE EMERGING EMPIRICAL FRONTIER

The explosion of commercial interest in artificial intelligence following the release of ChatGPT in late 2022 triggered a rapid emergence of corporate finance and accounting literature examining how public firms disclose AI (Eisfeldt, Schubert, and Zhang 2023; Babina et al. 2024; Basnet et al. 2025; Song et al. 2026; Anantharaman, Parker, and Rozario 2026). This literature builds upon the broader textual analysis tradition in finance (Li 2008; Loughran and McDonald 2011; Gentzkow, Kelly, and Taddy 2019), exploring how capital markets react to technological disclosure, how firms assemble AI-capable labor forces, and how managers communicate technological readiness to financial analysts and retail investors.

Simultaneously, a pressing regulatory and market concern has arisen: "AI washing," the corporate practice of exaggerating, fabricating, or opportunistically branding mundane software capabilities as cutting-edge artificial intelligence to inflate firm valuations or harvest a market premium (Gensler 2023; U.S. Securities and Exchange Commission 2024; Strauss et al. 2025). Borrowing conceptual scaffolding from the mature literature on greenwashing and corporate environmental decoupling

(Delmas and Burbano 2011; Lyon and Montgomery 2015; Marquis, Toffel, and Zhou 2016), researchers have begun testing whether corporate AI narratives represent substantive technological adoption or symbolic impression management (Song et al. 2026; Basnet et al. 2025; Anantharaman, Parker, and Rozario 2026).

1.3.2 FOUR GAPS IN THE CURRENT LITERATURE

Despite rapid progress, the existing literature on corporate AI disclosure and AI washing has four gaps:

1. **Counting words treats every mention as the same.** Most studies rely on keyword lists or simple term frequencies (counting “artificial intelligence,” “machine learning,” or “generative AI”) (Eisfeldt, Schubert, and Zhang 2023; Babina et al. 2024). That approach cannot tell an empty risk-factor sentence (“new AI tools developed by competitors could disrupt our market position”) apart from a specific claim about a named, deployed product with a measured result (“we deployed an internal predictive model that cut supply-chain logistics costs by 14%”). A word count cannot see who is making the claim, in what tense, at what stage, or whether it is promotional or operational.
2. **What firms disclose doing with AI is left unexamined.** Existing papers reduce corporate AI text to a single number: a 0/1 flag for “substantive mention” (Basnet et al. 2025), a sentiment score (Song et al. 2026), or an exposure index (Eisfeldt, Schubert, and Zhang 2023), and use it only as an input to a regression predicting stock returns, valuation, or analyst errors. Moving straight to that regression skips a more basic question: what do firms disclose doing with AI? Which business functions do they say they are automating: operations, marketing, customer service? Which technologies: predictive models, large language models, robotics? How much do they say is built in-house versus bought from a vendor? Are these described as live systems or early pilots?
3. **Studies look at one disclosure venue at a time.** Almost all existing work examines either annual filings (Form 10-K) or earnings calls, never both for the same firm-period. This misses the central tension in corporate communication: earnings calls are voluntary and unmonitored, where managers have every incentive to sound impressive (Frankel, Johnson, and Skinner 1999; Matsumoto, Pronk, and Roelofsen 2011), while 10-Ks and 10-Qs carry legal liability for material misstatement under Section 11 and Rule 10b-5. Without observing the same firm in both venues in the same period, prior work cannot measure the cross-venue gap that regulators have explicitly targeted in enforcement actions (Strauss et al. 2025).
4. **There is no benchmark for AI washing.** Greenwashing research can check corporate environmental claims against hard numbers (EPA toxic-release data, greenhouse-gas emissions) (Delmas and Burbano 2011; Marquis, Toffel, and Zhou 2016). AI adoption has no equivalent external census. Without one, AI-washing papers either treat washing as an unobservable regression residual or measure “talking without doing” against a benchmark that ignores how concrete and how grounded that talk’s own conduct already is.

1.3.3 WHAT THIS THESIS DOES ABOUT IT

This thesis addresses these four gaps in four steps:

1. **From keyword counts to structured claims.** Instead of counting words or computing hard-to-interpret text embeddings, the pipeline filters millions of paragraphs with an interpretable classifier, then has an LLM judge extract one structured claim per statement: who is speaking, what type of AI, in what tense (realized, planned, expected, hypothetical), about what business domain, using which concept from a fixed vocabulary, how specific, how promotional, and with what sentence-level evidence.

2. **An inventory of what firms say they do, not just a number for a regression.** Instead of reducing disclosure text to a single score, this thesis builds a descriptive inventory of 27,566 concrete activity claims (action, object, and business function) across 471 S&P 500 firms, recording the stage, audience, evidence, and technology source (in-house vs. third-party) behind each one. This is an inventory of what firms *say* they do, not an audited record of what they operationally do.
3. **Comparing the same firm across venues.** Using 479 firms that disclose in both venues across 2,281 firm-fiscal-year pairs, firm $\times$ fiscal-year fixed effects hold the firm and period fixed and compare only the call to the filing. The result is a within-firm, within-year venue gap: 10 times more promotional statements and 17 times more quantified claims per 1,000 words on the call than in the filing, for the same firm-year, a direct measurement of the venue mismatch that motivated recent SEC comment letters. This design removes everything common to the firm in that year; it cannot remove differences that come bundled with the venue itself (audience, format, legal exposure, topic selection all move together with venue). So the result is reported as a systematic venue-associated gap, not as the isolated causal effect of legal liability.
4. **A single, continuous AI-washing index.** Rather than an arbitrary tercile cut or an unobservable regression residual, this thesis defines AI washing as promotional excess conditional on a firm's own disclosed behaviour, the grounding (concreteness) of that behaviour, its overall AI-disclosure volume, and its document-type mix: a frame-level logistic benchmark predicts how promotional a firm should sound given those four things, and the standardized gap between observed and expected promotion is the index. Added to the economic test in Chapter 5 alongside disclosure volume, semantic content, and disclosed activity, the index adds essentially nothing to fundamentals-adjusted financial outcomes, which is itself the finding: promotional excess, once behaviour and grounding are already accounted for, is not the part of AI disclosure markets respond to.

1.4 THEORETICAL FOUNDATIONS

Three ideas from economics, accounting, and organizational sociology motivate the empirical designs that follow.

Signalling and cheap talk. Spence's (1973) signalling model says a disclosure is only credible if it is costly or creates legal exposure for a firm that is lying (Verrecchia 1983; Dye 1985). Crawford and Sobel (1982) show the opposite case: a costless, unverifiable statement carries no information in equilibrium. A generic AI claim ("we are leveraging AI across our enterprise") is cheap talk. A specific one (naming a model, citing a measured efficiency gain, naming a commercial partner) is costly, because it creates legal exposure if it turns out to be false (Rogers, Van Buskirk, and Zechman 2011). This is why the thesis separates promotional voice from specific, grounded activity throughout.

Institutional pressure to look current. Organizational sociology (Meyer and Rowan 1977; DiMaggio and Powell 1983; Suchman 1995) argues that under technological uncertainty, firms adopt the language and symbols expected of them to stay legitimate, whether or not that language reflects any change in what they actually do. This is a plausible reason why AI disclosure spread from 36% to 92% of the S&P 500 in four years, including in sectors where AI has limited operational relevance.

Talk that doesn't match conduct. Meyer and Rowan (1977) distinguish an organization's formal, ceremonial structure from what it actually does day to day; Westphal and Zajac (1994), Bromley and Powell (2012), and Crilly, Zollo, and Hansen (2012) extend this into different forms of that mismatch. In particular, Crilly et al. (2012) sharpen the theoretical distinction by contrasting intentional, opportunistic decoupling ("faking it", the precise institutional analogue of cosmetic AI washing) from

implementation friction (“muddling through”, reflecting legitimate managerial and technical delays in translating adoption plans into operational reality). In corporate disclosure, this shows up as impression management (Merkel-Davies and Brennan 2007; Huang, Teoh, and Zhang 2014): rhetoric that runs ahead of concrete execution. Throughout this thesis, that mismatch is measured directly by comparing a firm’s promotional voice to its disclosed operational behaviour.

Four terms used throughout, defined precisely in Chapter 4 (voice, disclosed behaviour, activity grounding, AI washing):

- **Voice (promotional posture):** the share of a firm’s AI statements written in promotional, strategic, or aspirational language.
- **Disclosed behaviour (disclosed activity volume):** measured from a separate LLM extraction pass over concrete, action-level claims (Chapter 2’s disclosed-activity inventory), kept deliberately apart from the frame-level posture measures used to build voice, so the two remain distinct instruments rather than restatements of the same content, though both still read the same underlying text, so any relationship between them is descriptive, not independently confirmed.
- **Activity grounding:** a firm-level score for how concretely disclosed activities are specified (named business function, deployed/scaled stage, named product, quantified outcome, named provider).
- **AI washing (the index):** promotional excess relative to what a firm’s own disclosed behaviour, grounding, volume, and document mix predict — a single continuous score, not a tercile cut. The related but distinct cross-venue gap (call vs. filing) is reported separately, in Chapter 6.

Propositions mapped to empirical research questions.

- *RQ1 (Disclosure Structure):* Tests whether disclosure volume and composition organize into a small number of stable, bootstrap-reproducible archetypes rather than a continuum of firm-level noise.
- *RQ2 (Decoupling):* Measures decoupling directly, by placing each firm on the voice-behaviour grid and testing how stable that placement is.
- *RQ3 (Economic Coherence):* Tests whether disclosure archetypes reflect economically meaningful differences in firm fundamentals or ceremonial legitimacy adoption, by examining those fundamentals after sector-year adjustment.
- *RQ4 (Signalling vs. Cheap Talk):* Tests whether specificity and concrete activities carry incremental economic information for asset pricing and firm fundamentals beyond disclosure volume and promotional tone.
- *RQ5 (Cross-Venue Disclosure):* Tests whether unmonitored voluntary venues (earnings calls) exhibit higher promotional inflation than legally monitored filings (10-Ks) for the same firm-year, a pattern consistent with, though not proof of, cheap talk in the unmonitored channel.
- *RQ6 (Regulatory Enforcement):* Tests whether SEC enforcement warnings functioned as a credible sanction capable of disciplining disclosure and closing the cross-venue gap.

1.5 CONTRIBUTION

The thesis makes three core contributions; everything else in the chapters that follow (the archetype taxonomy, the washing index, the SEC scrutiny check) builds on or synthesizes these three.

1. Measurement. A scalable, largely reproducible five-layer pipeline (prefilter, semantic frame, disclosed activity, firm-year aggregation, human validation) separates AI disclosure *volume* from its *semantic content* and its *disclosed activity*, recovering structured, falsifiable claims from 55,981 SEC documents and earnings-call transcripts at a cost and throughput a purely human-coded design could not reach. The pipeline is deterministic downstream of the two LLM extraction stages and documents its own measurement error by venue rather than assuming uniform accuracy.

2. Empirical finding: disclosed behaviour and specificity carry economic information; promotional excess does not. Beyond disclosure volume and financial fundamentals, the semantic content of what a firm says about AI adds 1.8–6.0 points of partial R^2 to beta, idiosyncratic volatility, price-to-sales, R&D-to-sales, and next-year revenue growth, rising to 2.1–7.8 points when disclosed activity is added alongside it. That signal is concentrated in specificity (named products, quantified outcomes) and in concrete disclosed activity, not in promotional register. Adding the AI-washing index (a firm-year residual, not a pooled firm effect — Chapter 4) on top of volume, content, and activity contributes nothing further on any of the five outcomes: R&D-to-sales, the outcome closest to a firm’s actual technological intensity ($\Delta R^2 = +0.001$, $p = 0.447$), beta ($p = 0.208$), idiosyncratic volatility ($p = 0.145$), price-to-sales ($p = 0.224$), or revenue growth ($p = 0.615$). Concrete disclosed behaviour and specificity are what markets respond to; a firm-year’s residual promotional excess, once behaviour and grounding are already priced in, is not.

3. Institutional finding: the same firm describes similar AI activity across venues, but attaches far more concrete evidence and promotion to it in earnings calls than in statutory filings. Comparing the same firm in the same fiscal year, the *mix* of disclosed activities is nearly identical between calls and filings: the difference is what evidence and register accompany the same claim. Calls add named products, quantified outcomes, and named providers/partners that the filing typically omits, alongside 10 times more promotional statements and 17 times more quantified claims per 1,000 words; the gap has widened since fiscal 2023 and shows no sign of closing after the SEC’s 2024 AI-washing enforcement action. This is a distinct, separately reported institutional result, not folded into the washing index.

Managerial contribution. For issuers, the thesis converts the measurement into a concrete checklist for credible disclosure (named activity, clear stage, quantified outcome, provider transparency, venue consistency) and quantifies a venue gap regulators are already scrutinizing. For investors, it identifies which textual dimensions carry incremental information (specificity and disclosed activity, not promotional tone) and shows that cross-venue consistency between the call and the filing is itself a usable credibility signal.

1.6 THESIS ROADMAP

- **Chapter 2** documents the corpus, the measurement pipeline, and its human validation.
- **Chapter 3** asks what firms disclose about AI: how disclosure grew and changed in composition 2021–2025, what activities firms describe doing, and whose technology they say it is.
- **Chapter 4** asks how firms differ: six conceptually defined dimensions of AI disclosure posture, the archetypes built on them, the single continuous AI-washing index (promotional excess conditional on disclosed behaviour, grounding, volume, and document mix), and whether these disclosure profiles correspond to economically distinct firms.
- **Chapter 5** asks whether the content of disclosure, beyond how much of it there is, carries incremental economic information.

- **Chapter 6** compares the same firm across venues and over time, including a check of whether the SEC's 2024 scrutiny produced a detectable break.
- **Chapter 7** draws out issuer and investor implications of the AI-washing index built in Chapter 4.
- **Chapter 8** concludes.

Table 1: Theory to empirical test.

Theory	What is measured	Hypothesis / RQ	Empirical test	Chapter
Institutional diffusion / legitimacy (DiMaggio and Powell 1983; Suchman 1995)	Share of firms disclosing, "no AI" segment	RQ6	Evolution 2021–2025, cohort check	3
Recurrent disclosure archetypes (Hennig 2007)	Six conceptually named disclosure-posture dimensions, then K-means on those dimensions	RQ1	K-means, bootstrap Jaccard stability, PCA robustness check	4
Decoupling (talk vs. behaviour) (Meyer and Rowan 1977; Crilly, Zollo, and Hansen 2012; Bromley and Powell 2012)	Promotional posture (frame-level) vs. disclosed activity volume (independent extraction)	RQ2	Continuous scatter + OLS residual (promotional excess)	4
Economic coherence of archetypes	Segment × sector-year residual profiles	RQ3	Sector-year residualized regressions	4
Signalling / cheap talk (Spence 1973; Crawford and Sobel 1982)	Specificity vs. promotional rhetoric	RQ4	Nested regressions, ΔR^2 , Wald, permutation	5
Cheap talk across venues (monitored vs. unmonitored) (Frankel, Johnson, and Skinner 1999; Matsumoto, Pronk, and Roelofsens 2011)	Call – filing promotional gap	RQ5	Firm × fiscal-year fixed effects, event study	6
Enforcement credibility (Gensler 2023; U.S. Securities and Exchange Commission 2024; Callaway, Goodman-Bacon, and Sant'Anna 2024)	Pre/post SEC scrutiny (DeepSeek shock as a secondary check, Appendix E)	RQ6	Event study with continuous exposure, DiD	6
Washing as a family of divergences (Delmas and Burbano 2011; Song et al. 2026)	Absolute mismatch, conditional excess, cross-venue gap	RQ2, RQ5	Three independent scores, cross-validation	7

2. DATA AND MEASUREMENT

2.1 CORPUS AND SOURCES

The U.S. corpus covers 10-K, 10-Q, DEF 14A, and 8-K filings for the S&P 500 universe plus earnings-call transcripts, from 2021 through a partial 2026 ([Table 2](#)). It comprises 55,981 documents (2,873 10-Ks, 8,099 10-Qs, 2,646 DEF 14As, 34,416 8-Ks, and 7,947 call transcripts) segmented into 9,138,778 paragraph instances. After a length-and-content scorability filter, 8,354,547 scorable instances collapse to 4,799,488 unique texts once repeated boilerplate is deduplicated by content hash. The full corpus, including the Italian and Chilean comparative panels, holds 10,216,373 paragraphs and 5,616,337 unique scorable texts; only the U.S. instrument carries embeddings, prefilter scores, and semantic frames, and is the basis for every result in this thesis.

Table 2: Documents by form and filing year.

Form	2021	2022	2023	2024	2025	2026	Total
10-K	492	496	490	490	490	440	2,898
10-Q	1,486	1,479	1,467	1,473	1,475	983	8,363
DEF 14A	494	495	487	483	492	432	2,883
8-K	6,679	6,202	6,120	6,142	6,288	4,398	35,829
Earnings call	1,874	1,865	1,839	1,795	574	0	7,947

The measurement pipeline moves through four stages: the raw corpus, a machine-learning prefilter, semantic-frame extraction, and disclosed-activity extraction. The unit of analysis changes at each stage (document, deduplicated text, individual claim, concrete activity) because each stage needs a different unit to do its job. Conceptually, this multi-stage inference architecture instantiates the classical Knowledge Discovery in Databases (KDD) process (Fayyad, Piatetsky-Shapiro, and Smyth 1996)—progressing through selection, preprocessing, transformation, data mining, and pattern evaluation—and aligns with the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Shearer 2000), mapping regulatory and corporate understanding into disciplined, reproducible data engineering and model evaluation stages.

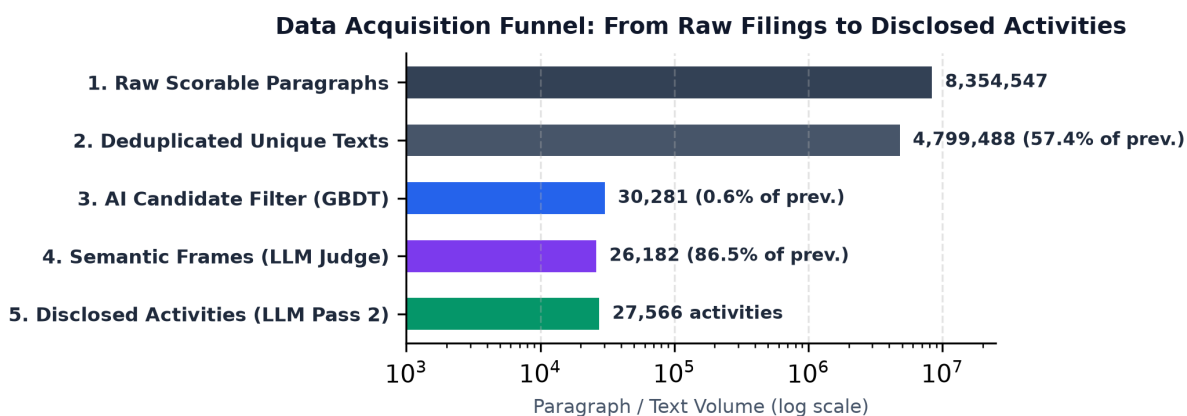


Figure 1: The Data Acquisition Funnel: progression of corpus volume across analytical screening stages (log scale).

Table 3: Compact summary of the inference funnel.

Stage	Unit of Observation	Operational Filter	Retained Volume	Pipeline Share
1. Raw Corpus	Paragraph instances	Universe coverage (S&P 500, 2021-2026)	8,354,547	100.00%
2. Deduplication	Unique scorable texts	BLAKE2b hash (> 3 char, alphanumeric)	4,799,488	57.45%
3. Machine Learning Gate	Candidate paragraphs	Supervised GBDT classifier (tau=0.17)	30,281	0.63%
4. Semantic Frames	Texts with >=1 frame	Zero-shot LLM judge (qwen3.7-flash)	26,182	0.55%
5. Disclosed Activities	Behavioral action triples	Firm-conditioned LLM extraction (pass 2)	27,566	0.57%

Notes: High-level overview of the five-stage inference pipeline. Granular form-by-form decompositions (10-K, 10-Q, DEF 14A, 8-K, and Earnings Calls) and intermediate model acceptance diagnostics are reported in [Table 23](#) (Appendix A).

2.1.1 HOW THE PIPELINE WORKS

Three design choices matter for interpreting the results that follow:

1. **Deduplication removes boilerplate before it can distort anything.** Corporate filings repeat the same boilerplate paragraph across quarters (Li 2008; Loughran and McDonald 2011).

Every scorable paragraph is hashed, which collapses 8.35 million paragraph instances into 4.87 million unique texts; scoring and LLM extraction run once per unique text, and results are then joined back to every document that repeats it. This keeps a recycled legal disclaimer from being counted, and scored, more than once.

2. **The LLM judge doubles as a second filter, not just an extractor.** The machine-learning prefilter is tuned for high recall (≥ 0.92) so it does not miss rare AI mentions in non-technical sectors, which means it also lets through some false positives. The LLM judge catches most of these: 13.4% of prefilter-flagged texts (4,110 of them) get zero semantic frames and are dropped, rising to 17.5% in earnings calls and 39.6% in Form 8-K, the noisiest venues.
3. **The model is told whose filing it is reading.** When extracting disclosed activities, the prompt includes the firm's name and ticker, so the model can tell a firm's own product from a vendor's. AMD's "Instinct accelerators" are coded as its own product (own); an insurance firm's use of NVIDIA GPUs and OpenAI APIs is coded as third-party (third_party).

2.2 UNITS OF ANALYSIS

The pipeline moves across seven nested units, each used where it is the natural denominator:

- the unique text (a deduplicated paragraph, the unit of scoring and classification);
- the semantic frame (one claim extracted from a text; a text can yield zero, one, or several frames);
- the disclosed activity (one action-object-function triple extracted from a behavioural frame);
- the document (the unit of intensity, since a text's frames are joined back to every document instance in which it appears);
- the firm-year (the unit of the economic analysis);
- the pooled firm (the unit of the cross-sectional segmentation and washing score); and
- the firm $\times$ fiscal-year $\times$ venue cell (the unit of the cross-venue comparison in Chapter 6).

A firm-quarter unit is used only for the SEC and DeepSeek event studies in Chapter 6, where calendar-quarter alignment is required to define a clean pre/post cut.

2.3 AI PREFILTER

Before any LLM reads a paragraph, a cheaper classifier decides whether it is worth reading at all. This gradient-boosted model looks at 39 signals in four groups:

- keywords (a hand-built list of AI terms and roughly 60 named entities);
- meaning (how close the text is to known AI language, using sentence embeddings);
- shape (length, capitalization, position in the document); and
- sentence-level cues.

It is trained on a 9,900-text golden set labeled by two independent LLM judges ([Table 6](#)) and scores 0.96 on the standard F1 metric for 10-K filings, meaning it rarely misses a real AI mention and rarely flags something irrelevant, on the document type it was tuned for. No part of this label set is human-annotated; that distinction, and what it means for how much to trust these numbers, is addressed directly in the Conclusions.

It does not do equally well everywhere. On a held-out sample never used in training, precision (the share of flagged texts that are actually about AI) stays at 0.98 for 10-K/10-Q, drops to 0.73 for proxy

statements and 8-Ks, and drops further to 0.59 for earnings calls. Recall (the share of true AI mentions the filter catches) stays above 0.92 everywhere, so the filter almost never misses something real; it just lets through more noise on some document types than others.

In total, it flags 30,281 texts as AI-relevant, 0.6% of everything scored. Every downstream number in this thesis counts a document with zero flagged texts as a documented zero, not as a missing observation: a firm that never mentions AI is data, not a gap in the sample.

2.4 SEMANTIC FRAME EXTRACTION

Every text the prefilter flags is then read once by an LLM (prior computational social science work shows LLMs match or beat human annotators on tasks like this (Gilardi, Alizadeh, and Kubli 2023; Ziems et al. 2024)), which pulls out zero, one, or several separate claims about AI, a “frame.” Each frame records eight things about the claim:

- who it’s about (the firm itself, a customer, a competitor, a supplier);
- what kind of AI (generative vs. predictive/ML: generative went from roughly half of mentions in 2021 to 93% by 2026);
- its tense (already happening, planned, expected, or hypothetical: 72% describe something already happening);
- its business domain;
- which concept from a fixed list it matches (adoption, capability, risk, or governance);
- how specific it is;
- whether it reads as promotional; and
- what evidence backs it.

This step also acts as a second, more careful filter: 13% of what the cheap prefilter flagged turns out, on a closer read, to have zero real AI claims and gets dropped. In total, 30,281 texts are read this way, yielding 43,366 distinct claims.

Table 4: Semantic frame schema: the eight fields recorded per claim, with observed value distributions.

Field	Description	Observed values
subject	Who the claim is about	firm (84%); competitors_or_industry (8%); customers (7%); suppliers_or_partners (1%)
ai_type	Generative vs. predictive/ML	unspecified (57%); generative (37%); predictive_ml (6%)
temporal	Tense: realized, planned, expected, hypothetical	realized (71%); planned (14%); expected (10%); hypothetical (5%)
domain	Internal, customer-facing, or unspecified	customer_facing (44%); internal (43%); unspecified (13%)
concepts	Fixed vocabulary of adoption/capability/risk/governance concepts (multi-label)	see Chapter 3 frame-type table
specificity	5 sub-flags: process, product, vendor, metric, date/timeline named	see specificity_index, Chapter 4
rhetoric	Promotional or strategic-importance framing (boolean)	14.9% promotional (corpus mean)
evidence	Sentence indices supporting the claim	always non-empty by construction

2.5 DISCLOSED ACTIVITY EXTRACTION

Whenever a claim describes the firm itself doing something concrete (deploying, investing, hiring, measuring a result), it goes through a second pass. This time the model is told which company it's reading, so it can tell "our own product" apart from "a tool we bought." For each such claim it records:

- the action (deploy, develop, partner, invest in infrastructure, hire, pilot, scale, measure a result, govern, restrict, and a few others);
- what the action targets (in free text, later grouped into families) and who benefits from it (employees, customers, internal process, partners);
- how mature it is (exploring, piloting, deployed, scaled);
- whose technology it is (the firm's own, a third party's, co-developed, acquired, open-source);
- who else gets named (a partner, a vendor, a competitor, a customer); and
- how solid the evidence is (a named product, a number, a named vendor, or just generic language).

Every one of these records is tied to an actual sentence; nothing is inferred without a supporting quote. This pass produces 27,566 distinct activities across 471 of the 510 firms.

Table 5: Disclosed activity schema: the fields recorded per activity, with observed value distributions.

Field	Description	Observed values
action	What the firm did	deploy (63%); invest_infrastructure (8%); scale (7%); pilot_or_explore (4%); ...
stage	Operational maturity	deployed (58%); scaled (14%); piloting (10%); exploring (9%); ...
ai_source	Whose technology it is	own (73%); unspecified (16%); third_party (8%); acquired (1%); ...
target	Who benefits	customers (48%); internal_process (25%); unspecified (14%); employees (9%); ...
evidence_strength	How solid the evidence is	generic (53%); named_product_or_process (29%); metric (13%); vendor (4%)
object	Free text, later grouped into 21 object families	see Chapter 3 action x object table
named_entities	Own brand, external provider/model, partner, acquired company, customer, distribution channel, competitor	17,193 entity mentions total

2.6 AGGREGATION

The measurement pipeline distinguishes between two complementary concepts of disclosure volume:

1. **Diffusion:** the share of filings (or of firms) with at least one AI statement. Bounded between 0 and 1, and unaffected by how many separate claims a document makes.
2. **Cumulative frame density:** the total number of AI statements per 1,000 words of scorable narrative text: how much AI content there is, not just whether it's present. Every "per 1,000" intensity figure in this thesis, including in the clustering, regression, and event-study analyses, uses words as the denominator rather than paragraphs, so a single unit of measurement holds throughout.

The two move together closely (a firm with more AI content is almost always also a firm that discloses at all), so it makes little difference which one a ranking or a regression uses. On average, a paragraph that discusses AI makes 1.44 separate statements, rising from 1.43 in 2021 to 1.49 in 2025 as filings got denser.

To go from a single document to a firm-year, each document's rate is combined with the others using the median, not an average weighted by length. The reason is practical: a firm-year can include one 10-K and several 10-Qs and 8-Ks of very different sizes, and the median stops one unusually long or short document from dominating the number. In practice this choice barely matters, because the 10-Q almost never contains an AI statement, so a firm-year's AI content usually comes from a single filing (its 10-K) anyway, and the median gives the same answer as a length-weighted average to the ninth decimal place across all 2,475 firm-years used in Chapter 5 (checked directly, not assumed).

For concepts describing behaviour (deployment, risk, governance, etc.), the default measure is a log-transformed count rather than a share of a firm's total AI statements: the count version turned out to be much more reliable when the same firm was checked against itself twice (a reliability score of 0.845 vs. 0.528 for shares).

Firm-level rates are also pulled slightly toward the corpus average when a firm has very few AI statements to base a rate on, so a firm with two mentions doesn't get treated as confidently as one with two hundred; a firm with zero mentions gets no rate at all and goes into its own "no AI" category instead.

Activity grounding, used in Chapter 4, is simpler: the average of five yes/no questions per activity:

- is the function named;
- is it deployed or scaled;
- is the product named;
- is there a number attached; and
- is the provider named.

2.7 MODEL VALIDATION AND TRAINING DATASETS

Every automated stage in the measurement pipeline is benchmarked against held-out ground truth and inter-annotator evaluation datasets. [Table 6](#) details the evaluation datasets, their sample design, and empirical performance metrics.

Table 6: Labeled evaluation datasets, validation benchmarks, and inter-judge agreement.

Benchmark Dataset	Forms	Sample Size	Annotated By	Metric / Empirical Performance	Status
Golden Set (Training)	10-K, 10-Q	9,900 texts	Qwen 3.7 / Gemini 3.8	F1 = 0.96 (Prec. 0.98, Rec. 0.94)	Verified
Inter-Judge Agreement	10-K, 10-Q	6,038 pairs	Gemini 3.8 vs. Qwen 3.7	Cohen's κ = 0.81–0.87 across dimensions	Verified
Out-of-Domain Proxy	DEF 14A, 8-K	1,500 texts	Qwen 3.7 Flash	Prec. 0.73, Rec. 0.92, F1 = 0.82 (14A), 0.65 (8-K)	Verified
Out-of-Domain Calls	Transcripts	800 texts	Qwen 3.7 Flash	Prec. 0.59, Rec. 0.98, F1 = 0.73 (Strong Prec. 0.85)	Verified
Lexical Anchors & Gate	All forms	27 anchors	Expert curated	39 signals, BGE-M3 cosine anchors	Verified
Activity Extraction	Mixed sample	30 paragraphs	Iterative manual inspection	Entity role & source prompt engineering	Verified

Benchmark Dataset	Forms	Sample Size	Annotated By	Metric / Empirical Performance	Status
Human Ground Truth	All forms	720 texts	Human coders (ui-validator/)	Human-to-LLM agreement (434 frames, 267 activities)	In progress

Notes: All operational labels supporting the empirical chapters derive from zero-shot LLM extraction. The measured inter-judge agreement ($\kappa = 0.81\text{--}0.87$) establishes strong reproducibility across distinct foundation model architectures (Google Gemini vs. Alibaba Qwen).

2.8 FINANCIAL AND MARKET VARIABLES

Firm-year financial and market variables are drawn from XBRL and price data, aligned by actual fiscal period end rather than filing date, with documented fallback chains for concept coverage (capex 87%, SG&A 80%, net margin 98%, revenue 98%) and a [340, 380]-day gap rule for validating year-over-year changes. The variable set used throughout Chapters 3–5 comprises market capitalization, R&D-to-sales, gross margin, operating margin, asset turnover, price-to-sales, next-fiscal-year revenue growth, market beta, and idiosyncratic volatility (the annualized standard deviation of one-factor market-model residuals over the 252 trading days preceding the filing, following Campbell et al. (2001) and Ang et al. (2006)). All financial ratios are winsorized at the 1st and 99th percentiles.

3. WHAT FIRMS DISCLOSE ABOUT AI

3.1 EVOLUTION 2021–2025

The expansion of AI disclosure can be tracked on two complementary margins: **diffusion** (the share of S&P 500 firms making at least one AI statement in their SEC filings) and **cumulative frame density** (AI statements per 1,000 words of scorable narrative text). Diffusion rises from 35.7% in 2021 to 92.1% in 2025. Concurrently, cumulative frame density rises from 0.016 to 0.111 frames per 1,000 words (a 6.9-fold increase).

The ratio of frames per AI-mentioning paragraph expands from 1.43 in 2021 to 1.49 in 2025, showing that as AI disclosure diffused, disclosing paragraphs also became semantically denser, frequently addressing risk, governance, and operational rollout in the same passage. The single largest discontinuity occurs in the 2024 filing season, when fiscal-2023 10-Ks (the first written with a full year of generative AI exposure) push the share of 10-Ks with any AI content from 39% to 72%. A balanced cohort of the 475 firms present in both 2021 and 2026 reproduces the identical trajectory of rising diffusion, confirming that market-wide diffusion rather than sample attrition drives the trend.

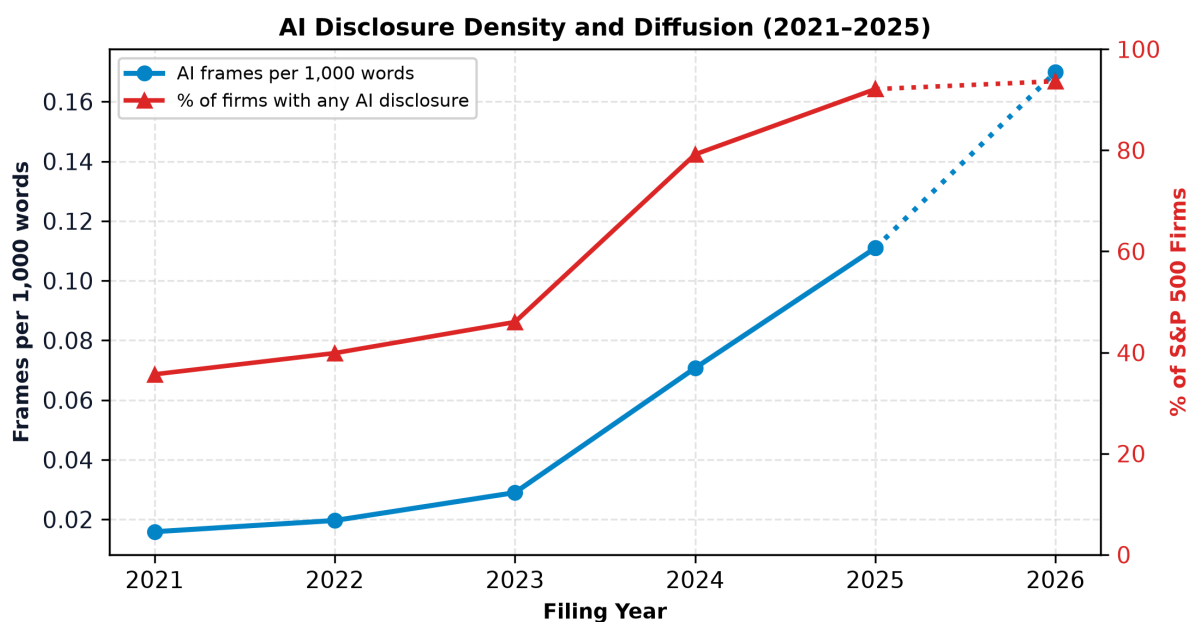


Figure 2: Evolution of AI disclosure density (AI statements per 1,000 words of scorable narrative text) alongside the share of disclosing S&P 500 firms, 2021–2025.

The composition of that growth matters more than its scale: risk grows roughly five times faster than promotion, and governance three times faster:

Table 7: Frame-type growth and composition, 2021-2025.

Frame type	Growth, 2021-2025	Share of firm's AI frames, 2021	Share, 2025
Risk	20.3x	11.4%	33.2%
Governance	11.7x	7.2%	12.1%
Realized-outcome	6.0x	80.8%	70.0%
Promotional	4.2x	13.7%	8.2%
Deployment	4.2x	51.1%	30.7%

Notes: Shares are of a firm's own AI frames and need not sum to 100%, since a single frame can carry more than one concept (e.g., a realized, risk-related statement).

What firms added to their filings between 2023 and 2025 was, above all, an AI risk factor and a governance line; the product pitch is being made somewhere else, a pattern the venue-gap analysis in Chapter 6 confirms directly.

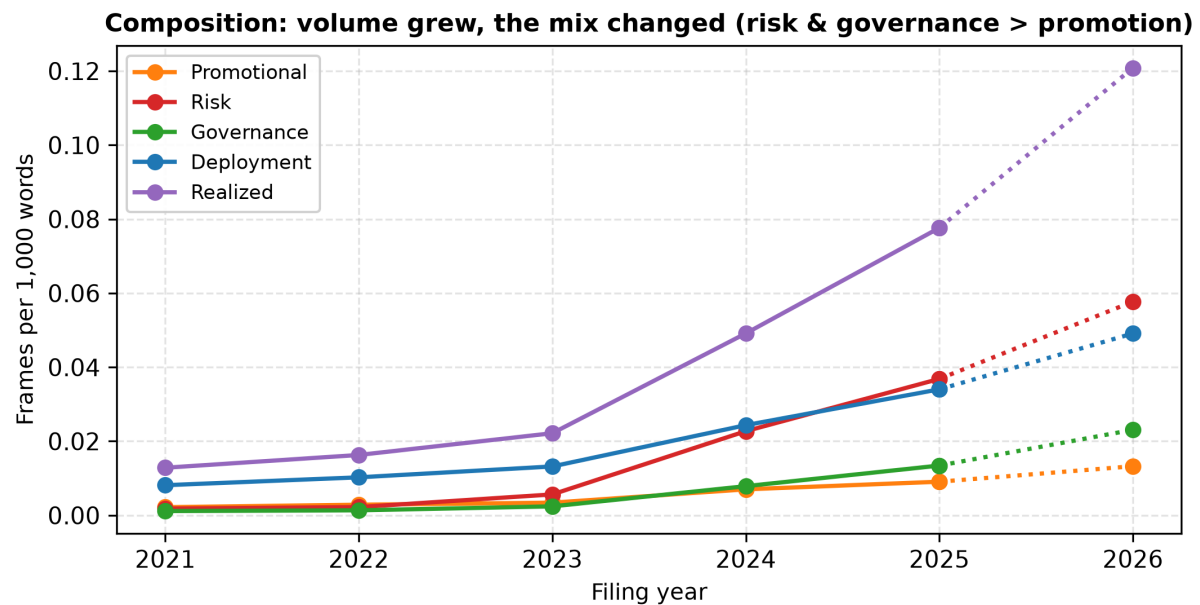


Figure 3: Evolution of semantic frame composition in corporate filings (2021–2025), showing the rapid rise of risk and governance disclosures relative to promotional claims.

3.2 WHAT FIRMS SAY ABOUT AI

Before any clustering, the raw distribution of frame concepts already separates quantity from composition:

- Realized framing dominates temporal claims (72.1%) over planned (12.5%), expected (8.2%), and hypothetical (7.2%) claims.
- The firm itself is the subject of 85.3% of statements, versus 9.1% for competitors and industry, 4.9% for customers, and 0.6% for suppliers and partners.
- Internal use narrowly exceeds customer-facing use in filings (46.6% vs. 37.7%), while the ratio inverts in earnings calls (35.1% internal vs. 55.6% customer-facing): the filing describes process, the call describes product.
- Cybersecurity, regulatory/legal exposure, and competitive disruption dominate the risk vocabulary; generative AI overtakes predictive/ML mentions between 2022 and 2023 and reaches roughly 93% of type-declared mentions by 2026.

These raw distributions motivate, but do not by themselves establish, the archetype structure developed next.

What firms say about AI, before any segmentation

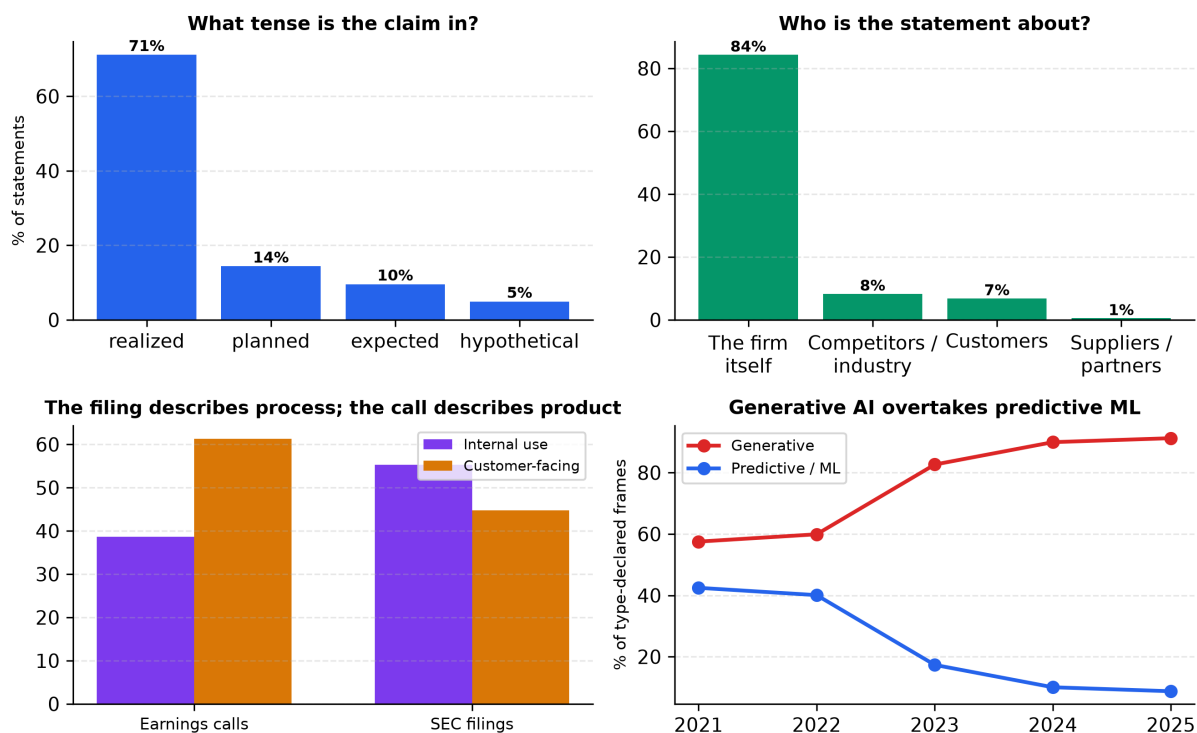


Figure 4: The raw semantic landscape before any clustering: temporal framing, who is speaking, internal vs. customer-facing use by venue, and the generative/predictive shift over time.

3.3 DESCRIPTIVE INVENTORY OF DISCLOSED CORPORATE AI ACTIVITIES

To see what public corporations literally disclose doing with artificial intelligence, rather than reducing that to a single number, the second-stage extraction pipeline recovers 27,566 unique disclosed activities across 471 S&P 500 firms (92.4% of firms with filings). Every activity represents an extracted (action, object, function) triple grounded in a verified sentence of textual evidence, annotated with its operational maturity stage, target audience, technology origin, and named entities.

3.3.1 AGGREGATE BEHAVIORAL LANDSCAPE, 2021–2025

Six dimensions describe what a firm is doing, how mature it is, for whom, whose technology it is, how well evidenced the claim is, and which business function it serves, and each of the six moved between 2021 and 2025, in the same direction: toward less maturity and less concreteness per activity, even as the number of activities disclosed each year grew several-fold.

- The deployed share of activities falls from 65% to 59%, while piloting and exploring combined rise from 16% to 18% (peaking at 25% in 2023).
- The own-technology share drifts down from 76% to 74% while third_party roughly doubles, from 6% to 8%.
- Claims backed by a named product or process fall from 34% to 30% while generic, unanchored language rises from 48% to 53%.

- The share of activities left with no identifiable business function climbs from 24% to 33%.

In short: the marginal activity added since the 2023 boom is, on average, less mature and less concretely evidenced than the ones disclosed before it. Volume grew faster than substance.

Disclosed AI activity got bigger in volume but not more concrete, 2021–2025

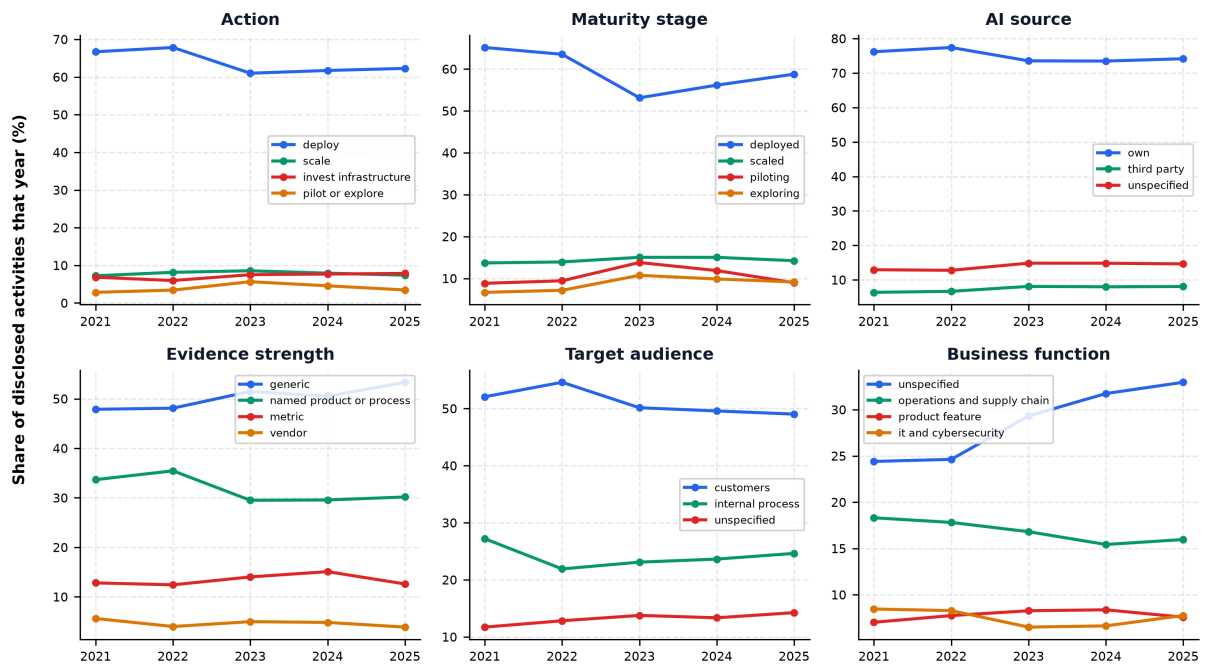


Figure 5: Evolution of disclosed AI activity across five dimensions, 2021–2025 (share of activities per category, by year).

[Table 24](#) (Appendix A) reports the pooled 2021–2026 total for each category (the single number a reader wants for “what share of activities are X, overall”); the figure above is the one that tells the story of how that pool got there.

3.3.2 WHAT FIRMS DISCLOSE DOING: ACTION × OBJECT TYPOLOGY

When free-text objects are standardized into technology families, an empirical typology of corporate deployment emerges: what most S&P 500 firms disclose is deploying an AI product, a predictive model, or a data platform, not building AI from scratch. The top 10 combinations by share of firms are shown below; the full ranking of 19 combinations, with literal textual examples, is in [Table 21](#) (Appendix A).

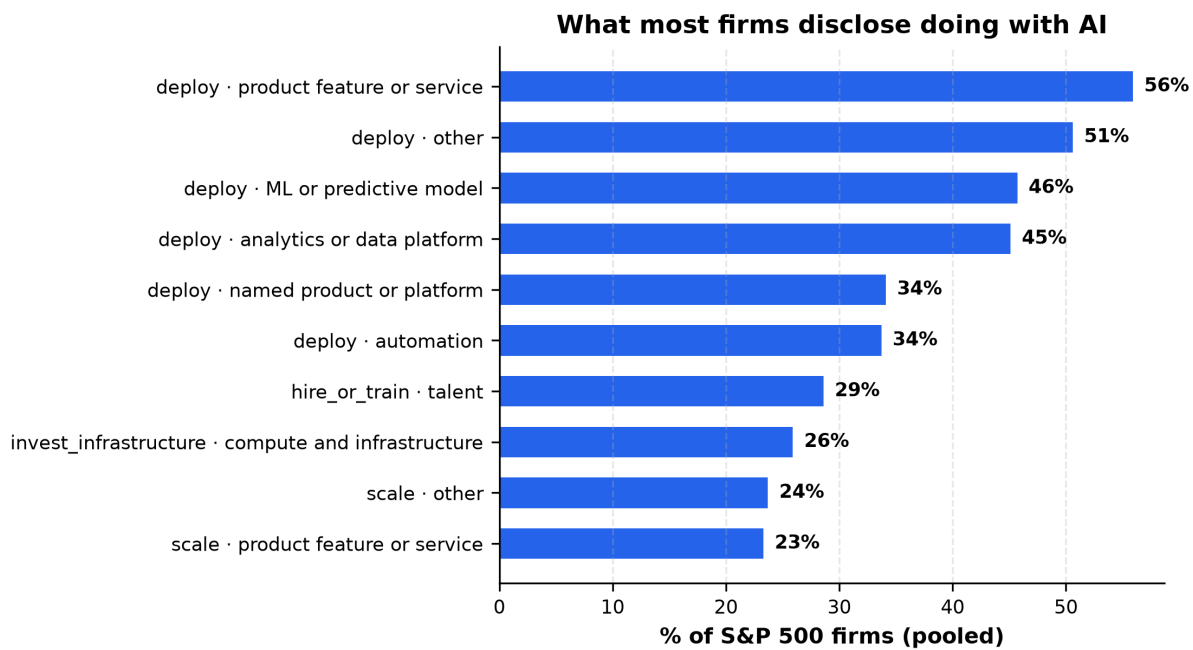


Figure 6: Top 10 action × object combinations by share of S&P 500 firms disclosing at least one such activity.

Intersecting actions, objects, and business functions reveals that the single most prevalent corporate application across the U.S. economy is deploying AI tools to optimize internal operations (30.0% of firms), automating workflows (27.0%), deploying predictive models in supply chains (19.0%), and establishing centralized operational data platforms (18.0%). Customer-facing conversational tools (17.5%) and embedding AI into core sellable products (17.0%) form a secondary cluster, while coding assistants (9.6%), marketing recommendation engines (7.6%), and fraud risk management (8.8%) occupy specialized vertical domains.

3.3.3 TECHNOLOGY PROVENANCE: PROPRIETARY BRANDING VS. VENDOR ATTRIBUTION

A central empirical question in corporate AI is whether public firms are genuine technology creators or commercial consumers of foundation models developed by hyperscalers. Across the corpus, the pipeline extracted 17,193 named entities, classifying each by its operational role.

The distribution of entity roles shows a strong proprietary branding bias:

Table 8: Named-entity roles across the corpus.

Entity role	Instances	Share
Own product or brand	13,333	77.5%
External technology vendor	1,185	6.9%
Strategic commercial partner	780	4.5%
Enterprise customer	728	4.2%
Acquired technology startup	551	3.2%
External foundation model	474	2.8%
Distribution channel	72	0.4%
Competitor or reference	70	0.4%

17,193 named entities total.

Overall, 246 of 510 firms (48.2%) disclose at least one external vendor, commercial model, or technology partner. [Table 9](#) details the external vendor ecosystem cited by S&P 500 corporations.

Table 9: External AI technology ecosystem cited by S&P 500 corporations.

External technology provider / model family	% of Firms citing	Literal named entities extracted
Specialized Enterprise & Data Vendors	41.8%	AMD, Palantir, ServiceNow, Hugging Face, Snowflake
OpenAI / Microsoft Ecosystem	17.3%	OpenAI, Microsoft, ChatGPT, Azure, GitHub Copilot
Google / Alphabet	9.6%	Google, Google Cloud, Gemini, Google AI, Google Vertex AI
NVIDIA	9.0%	NVIDIA, Nvidia, NVIDIA H100, NVIDIA AI software, NVIDIA H100s
Amazon / AWS	6.9%	AWS, Amazon Web Services, Amazon, Amazon AWS, Bedrock
Meta Platforms	3.5%	Meta, Llama, Llama 3, Meta's Llama 3.1, Meta's Llama 2
Anthropic	2.4%	Anthropic, Anthropic, PBC, Anthropic Claude, Anthropic Cloud, Claude
IBM	2.4%	IBM, Watson, Notre Dame — IBM Technology Ethics Lab, watsonx, ibm
Salesforce	1.4%	Salesforce, Salesforce Platform, Einstein

Notes: Vendor citations include entities tagged as `external_provider`, `external_model`, or `partner`. Entities appear in this table only when cited by *other* firms as an external technology source (e.g., Microsoft citing OpenAI, or Lowe's citing Palantir and NVIDIA), excluding self-promotional brand mentions.

Only 5% of all activity statements name an external vendor. Most disclosures leave the technology's origin unstated or attribute it to the firm itself.

While OpenAI and Microsoft represent the dominant external foundation model stack (cited by 17.3% of S&P 500 firms), external vendors are mentioned in only 5.0% of total activity statements. Fully 35.0% of all activity statements explicitly name a proprietary corporate brand. Public companies publicize *what* they achieve with artificial intelligence far more than the third-party infrastructure and foundational models behind those achievements.

3.3.4 SECTORAL DISTRIBUTION OF FRAMES AND ACTIVITIES

The two extraction passes described above (semantic frames and disclosed activities) are shown here purely descriptively, each on its own, crossed against sector (2-digit SIC). This is not yet the clustering of Chapter 4: no attempt is made here to relate one pass to the other, only to see where each one concentrates.

Table 10: Frame composition by sector (2-digit SIC, top 10 by firm count): median rates per 1,000 words.

Sector	Firms	Frames/1k	Promo/1k	Risk/1k	Gov/1k
Business services	66	0.196	0.024	0.054	0.02
Instruments	42	0.063	0.003	0.022	0.006
Chemicals	39	0.033	0.002	0.014	0.005
Electric/gas/sanitary	34	0.016	0	0.007	0.004
Electronic equipment	29	0.135	0.015	0.03	0.008
Industrial/computer equip.	28	0.141	0.017	0.029	0.011
Holding/investment offices	27	0.028	0.001	0.012	0.004
Insurance carriers	23	0.038	0.002	0.017	0.01
Depository institutions	21	0.024	0.001	0.01	0.005
Food products	18	0.017	0	0.009	0.003

Table 11: Disclosed activity volume and composition by sector (2-digit SIC, top 10 by firm count).

Sector	Firms	Median activities/firm	Proprietary tech share
Business services	66	101	79.8%
Instruments	42	23	68.0%
Chemicals	39	10	53.5%
Electric/gas/sanitary	34	7	35.1%
Electronic equipment	29	38	84.8%
Industrial/computer equip.	28	54	78.0%
Holding/investment offices	27	9	39.9%
Insurance carriers	23	19	55.6%
Depository institutions	21	16	52.7%
Food products	18	4	38.1%

The two tables are shown side by side deliberately, not merged: [Table 10](#) describes how each sector *talks*, [Table 11](#) describes what each sector *declares doing*, and the two tell related but distinct stories. Business Services and Industrial/Computer Equipment lead on both overall frame volume and activity count: the sectors doing the most AI-related disclosure also report the most concrete activity. But composition differs: risk and governance content together make up 65–69% of the (much smaller) frame volume in Insurance Carriers, Depository Institutions, and Electric/Gas/Sanitary, against 38% in Business Services: the more heavily regulated sectors disclose proportionally more defensively even where they disclose less overall, foreshadowing the governance-versus-conduct separation Chapter 4 develops formally.

4. DISCLOSURE ARCHETYPES AND THEIR ECONOMIC MEANING

4.1 SIX DIMENSIONS OF AI DISCLOSURE POSTURE

The rest of this chapter used to run two analyses that never had to agree with each other: a K-means over six raw disclosure features producing three archetypes, and a *separate* cut of promotional voice against disclosed behaviour into terciles producing a 3×3 grid.

A second problem sat underneath the first: three of those six clustering features (deployment, capability, adoption stage) are declared *actions*, the same kind of content the disclosed-activity extraction in Chapter 2 records on purpose, as a second, independent LLM pass, while the other three (risk, governance, promotion) are pure *communication style*. Mixing the two meant the clustering could never be independently validated against the activity inventory, because part of what it clustered on already *was* activity-adjacent content.

This chapter fixes both problems by keeping a strict separation the rest of the pipeline already makes: clustering uses only disclosure posture (how a firm talks about AI), and the disclosed-activity inventory is reserved entirely for characterizing the result afterward, never for building it. Because both extractions read the same corporate text, that characterization is descriptive, and the chapter says so explicitly where it matters (the “Cross-pipeline characterization” section below); the only genuinely external check in this chapter is against XBRL and market data, which never passed through the LLM pipeline at all (Economic distinctiveness, below).

Six posture dimensions are computed directly from gold_ai_frames, each a simple, named average of a small number of raw indicators, shrunk empirically toward the corpus mean in proportion to how many frames support it (Chapter 2's aggregation rule):

- **Promotional posture:** the average of promotional and strategic-importance rhetoric.
- **Risk orientation:** the share of statements carrying a risk concept.
- **Governance orientation:** the share carrying a governance concept.
- **Temporal posture:** the share of statements describing something already realized, rather than planned, expected, or hypothetical.
- **AI positioning:** the share framed as customer-facing rather than internal.
- **Specificity:** the average of five concreteness flags (named business process, product, vendor, quantified metric, date/timeline).

A seventh dimension, **disclosure intensity** (a percentile rank of AI statements per 1,000 words, to keep sample noise in a firm's word count from moving it across a cluster boundary), is included for the same reason the superseded design included it: a firm with three AI statements and one with three hundred, in the same register, are still not the same firm.

Table 12: Correlation matrix of the seven posture dimensions.

	Promotional	Risk	Governance	Temporal (realized)	Customer-facing	Specificity	Intensity
Promotional	1	-0.4	0	0.17	0.3	0.27	0.31
Risk	-0.4	1	-0.12	-0.48	-0.48	-0.58	-0.37
Governance	0	-0.12	1	0.24	-0.35	-0.07	-0.2
Temporal (realized)	0.17	-0.48	0.24	1	0.27	0.43	0.21
Customer-facing	0.3	-0.48	-0.35	0.27	1	0.61	0.6
Specificity	0.27	-0.58	-0.07	0.43	0.61	1	0.39
Intensity	0.31	-0.37	-0.2	0.21	0.6	0.39	1

Governance orientation is close to uncorrelated with everything else ($|r| \leq 0.35$, mostly under 0.20): a firm's governance framing says little about its risk framing ($r = -0.12$), its promotional register ($r = 0.00$), or how much it discloses ($r = -0.20$). Risk orientation, by contrast, is the most connected dimension: it correlates negatively with realized/temporal posture (-0.48), specificity (-0.58), customer-facing positioning (-0.48), and intensity (-0.37): a firm that frames AI mostly as a hypothetical risk tends to say less about it, less concretely, and less often about a customer-facing use, all at once.

How this space is diagnosed, not built, by PCA. A principal-components analysis of these seven standardized dimensions is reported as a robustness check, not as the basis for the clustering below: if the first two components captured most of the variance, that would argue for collapsing the space to a two-dimensional summary instead of clustering on all seven.

Table 13: PCA diagnostic on the seven posture dimensions: variance explained, reported as a robustness check, not as the clustering method.

Component	Variance explained	Cumulative
PC1	43.2%	43.2%
PC2	20.2%	63.3%
PC3	12.1%	75.4%
PC4	8.4%	83.8%
PC5	7.0%	90.8%

Component	Variance explained	Cumulative
PC6	5.4%	96.1%
PC7	3.9%	100.0%

The first two components explain 63.3% of the variance (43.2% and 20.2% respectively) and the third still adds a non-trivial 12.1%, not a dominant low-dimensional structure that would justify replacing the seven named dimensions with a two- or three-component summary, but also not so flat that the dimensions are pure noise. This supports the chapter's design choice: keep the six conceptually distinct, managerially interpretable posture dimensions (plus intensity) for clustering, rather than substitute an abstract linear combination whose loadings would need their own translation back into plain language.

4.2 DISCLOSURE ARCHETYPES

Firms that never mention AI in their SEC filings get their own “No AI” label by rule: there is no “how it talks” to measure for a firm that says nothing. For everyone else, K-means clusters firms on the seven standardized posture dimensions above, and the number of clusters is chosen by resampling each firm's own frames, refitting the entire pipeline (posture rates and clusters) on the resample, and keeping the largest k whose worst-recovered cluster still clears a 0.60 mean-Jaccard floor against the original assignment.

Table 14: Choosing k by bootstrap stability: mean Jaccard overlap per cluster (25 resamples of each firm's own frames).

k	Jaccard by cluster	Minimum	vs. 0.60 floor
2	0.82, 0.90	0.82	usable
3	0.66, 0.76, 0.81	0.66	usable
4	0.73, 0.63, 0.57, 0.71	0.57	below floor
5	0.44, 0.53, 0.70, 0.61, 0.67	0.44	below floor

k = 3 is the largest value clearing the floor (k = 4 and k = 5 both fail, driven by at least one unstable cluster each).

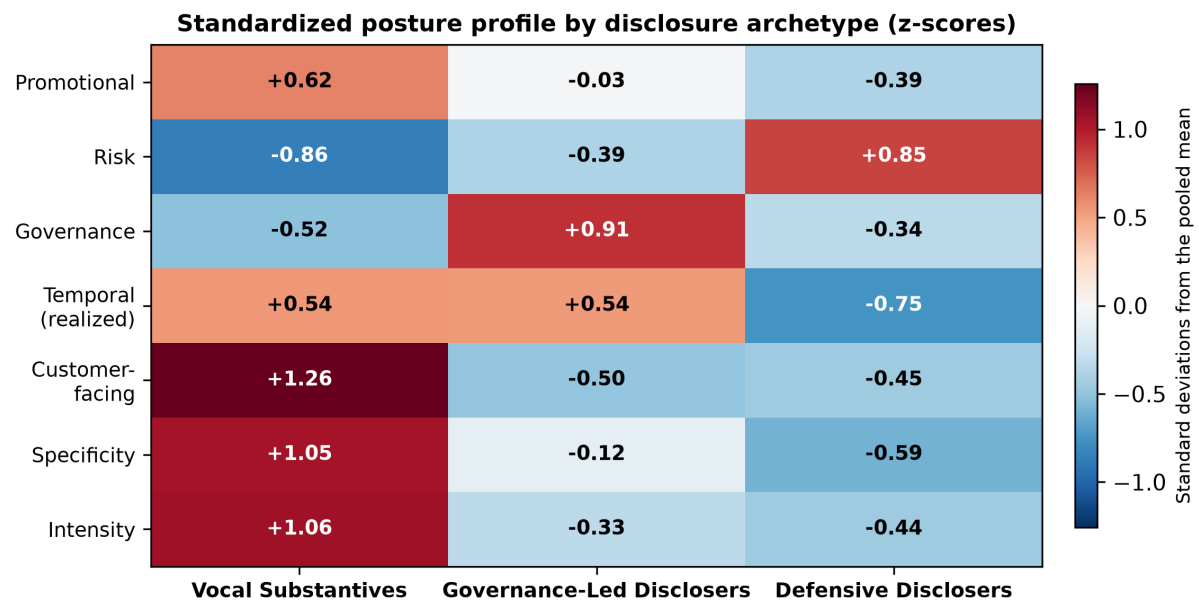


Figure 7: Disclosure archetypes at a glance: standardized (z-score) mean posture profile of the three K-means archetypes.

The three archetypes plus “No AI.” Across the pooled cross-section of 510 firms: **Vocal Substantives** (124 firms, stability 0.76: the highest promotional register, the highest customer-facing positioning, the highest specificity, and the highest disclosure intensity of the three), **Governance-Led Disclosers** (142 firms, stability 0.81: the only archetype defined by above-average governance orientation, with moderate everything else), and **Defensive Disclosers** (191 firms, stability 0.66, the largest archetype: the highest risk orientation and the lowest temporal-realized, customer-facing, specificity, and intensity scores), plus **No AI** (53 pooled firms with fewer than five AI statements in their SEC filings across the sample period). Year-over-year archetype persistence in the panel is 69.3%.

4.2.1 CROSS-PIPELINE CHARACTERIZATION: WHAT THE ARCHETYPES REPORT DOING

None of the three archetypes above was built from any declared-action content: the clustering runs entirely on frame-level communication posture. The disclosed-activity inventory (a separate LLM extraction pass, Chapter 2) is used here to characterize what each archetype reports doing.

This is **not independent validation**: both extractions read the same underlying corporate text, so a firm that discloses more about AI overall has mechanically more opportunity for the semantic-frame pass to find promotional or governance language *and* for the disclosed-activity pass to find a concrete action. A firm’s total AI disclosure volume can drive both sides of the comparison at once, without promotional register or governance framing being substantively related to conduct at all. The activity layer is therefore reported here as a descriptive, cross-pipeline characterization, and the comparison below conditions on total AI disclosure volume (log frame count) to reduce this mechanical co-movement before reading anything into the remaining difference.

Table 15: Disclosed behavioral profile by disclosure archetype, from the disclosed-activity inventory (never an input to the posture clustering).

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Disclosed activities per firm (median)	96	18	8	1
Customer-facing activity share (%)	53.6%	30.1%	33.1%	33.5%
Technology source: proprietary/own (%)	80.0%	51.7%	50.6%	40.8%

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Activities naming an own product/brand (%)	40.7%	17.2%	16.7%	18.3%
Activities reporting a quantified outcome (%)	15.2%	10.4%	10.7%	13.6%
Activities at stage deployed/scaled (%)	74.7%	68.7%	59.5%	58.6%

Notes: Distributions computed from the 27,566-activity inventory across 510 S&P 500 firms, joined post hoc to the posture-based archetype assignment. “No AI” firms with positive activity counts reflect conference-call statements by firms whose formal SEC filings omit AI entirely.

The raw medians in [Table 15](#) are exactly the pattern a shared-source-text confound would produce on its own: the archetype with the most total AI disclosure (Vocal Substantives, Table “Disclosure archetypes at a glance”) also reports the most activities. The question worth asking is whether any of that difference survives once total disclosure volume is held fixed.

Table 16: Disclosed activity count by archetype, before and after conditioning on total AI disclosure volume (log frame count). Reference category: Defensive Disclosers.

Archetype vs. Defensive Disclosers	Raw ratio	p	Volume-adjusted ratio	p
Governance-Led Disclosers	1.77x	<0.001	1.44x	<0.001
Vocal Substantives	11.24x	<0.001	2.39x	<0.001
log(1 + AI frames)	-	-	+1.00 (coef.)	<0.001

Total AI disclosure volume explains most of the raw gap (log frame count alone reaches $R^2 = 0.74$ in the adjusted model), exactly as the shared-source-text concern predicts. But a real difference survives conditioning on it: at the same total disclosure volume, Vocal Substantives still report 2.4 times as many activities as Defensive Disclosers ($p < 0.001$) and Governance-Led Disclosers report 1.4 times as many ($p < 0.001$). This is read as a **descriptive cross-pipeline characterization, not independent validation**: the composition statistics in [Table 15](#) (proprietary technology, customer-facing share, deployed/scaled stage) come from the same corpus and carry the same caveat, but the volume-adjusted gap is evidence the three posture archetypes are not simply relabeling “how much a firm discloses.”

- **Vocal Substantives** report 80.0% of activities claiming proprietary technology, 40.7% naming a specific own product or brand, 53.6% targeting customers directly, and 74.7% reaching a deployed or scaled stage.
- **Governance-Led Disclosers** show a composition close to Defensive Disclosers (51.7% proprietary, 30.1% customer-facing) despite reporting more activities at the same disclosure volume: governance orientation looks like a distinct communication register layered on top of a similar underlying activity mix, not a proxy for either more or less operational grounding.
- **Defensive Disclosers**, the largest archetype, show the lowest deployed/scaled share (59.5%) of the three and the lowest volume-adjusted activity count, consistent with the risk-heavy, low-specificity register documented in the posture profile above.

4.3 MAPPING THE ARCHETYPES: PROMOTIONAL POSTURE AGAINST DISCLOSED CONDUCT

The superseded design cut promotional “voice” and disclosed “behaviour” into terciles to build a separate 3×3 grid, a second, unrelated instrument on top of the archetypes above, and one that built its “behaviour” axis from the same frame-level concepts the clustering also used.

This section replaces that grid with a single continuous scatter, again carrying the same caveat as the cross-pipeline characterization above: **promotional posture** (from the frame-level posture clustering) on the y-axis, against **disclosed activity volume** (from the disclosed-activity extraction pass) on the x-axis, both read off the same underlying corporate text. Because promotional posture is a *rate* (a share of a firm’s own frames), rather than a raw count, it is less mechanically tied to how much a firm discloses overall than a raw activity count is, and the check below confirms the relationship is not simply a volume artifact. Still, the result is reported as a descriptive relationship between two measurements of the same corpus, not as independent confirmation.

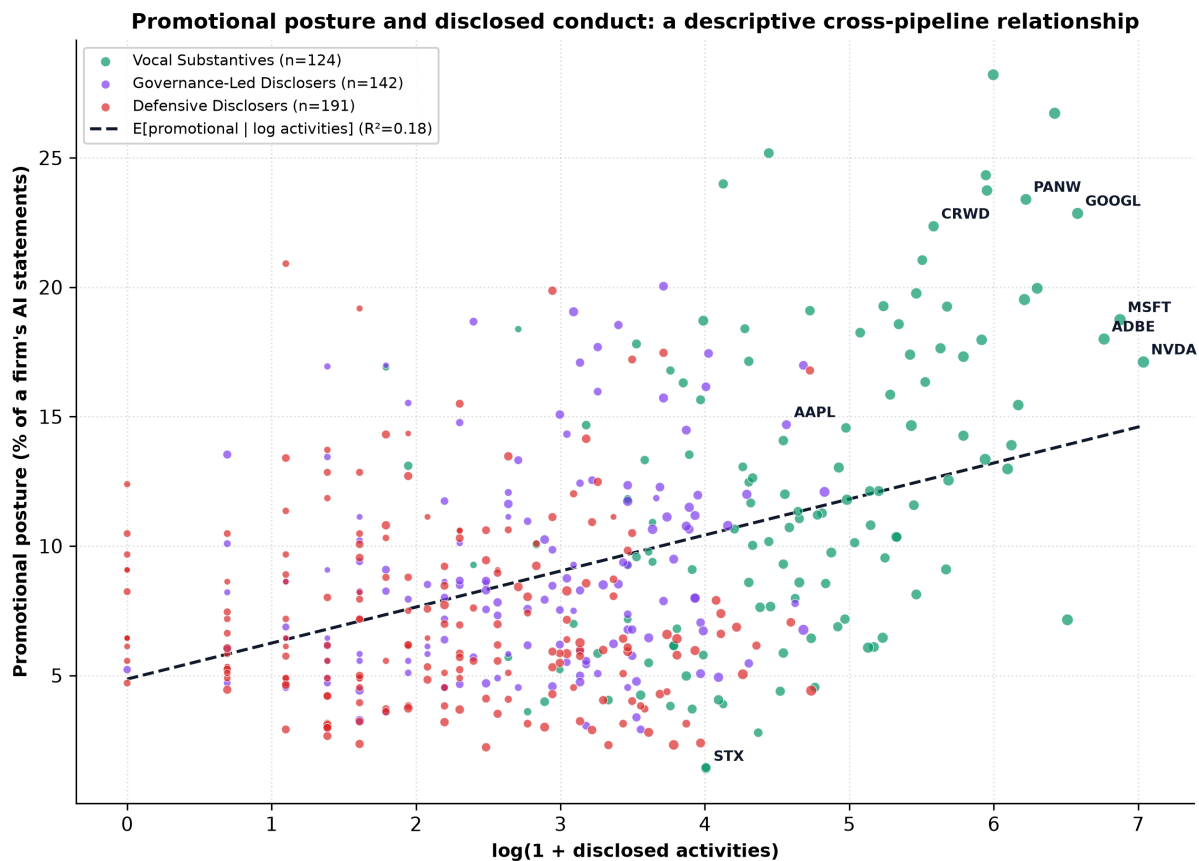


Figure 8: Promotional posture vs. disclosed activity volume (log scale), colored by disclosure archetype. The dashed line is the fitted $\text{promotional_posture} \sim \log(1 + \text{activities})$ relationship; vertical distance from it is a firm’s promotional excess. Both axes are read off the same underlying corporate text (semantic frames and disclosed activities are two LLM passes over the same documents), so this is a descriptive relationship, not an independent test.

Promotional posture rises with disclosed activity volume ($\beta = 0.014$ per log-activity unit, $p < 0.001$, $R^2 = 0.176$, $n = 457$). Adding total AI frame count as a control leaves this essentially unchanged ($\beta = 0.012$, still $p < 0.001$, while frame count itself is not significant, $p = 0.32$), so the relationship is not

simply “firms with more text score higher on everything,” even though both measurements come from the same corpus.

Firms that disclose more concrete AI conduct also, on average, promote it more, which is consistent with no distinctly “promotional” archetype emerging above: promotional posture is not a separate population of firms, it is a register that scales with disclosed conduct.

Promotional excess, the vertical residual from the fitted line, is a continuous score for how much more (or less) promotional a firm is than its own disclosed activity volume predicts, replacing the old grid’s single high-voice/low-behaviour corner with a quantity every firm has, not just the ones in one cell. Like the rest of this section, it should be read as a within-corpus descriptive score, not as an externally validated measure. The rest of this section formalizes that residual into the thesis’s AI-washing index: the same idea, but modeled at the frame level with more than one covariate.

4.4 THE AI-WASHING INDEX

This thesis defines AI washing as a single, continuous quantity: **promotional excess conditional on disclosed behaviour** — how much more promotional a firm’s language is than what its own disclosed conduct, the concreteness (grounding) of that conduct, its overall AI-disclosure volume, and the mix of document types it discloses through would predict, at the firm-year level, within sector and year.

An earlier version of this index fitted a logistic model on individual frames, pooled across all five years of a firm’s disclosure into one number per firm. Two things were wrong with that design, not just its presentation. First, pooling by firm meant the score never varied within a firm across years, which throws away the only evidence that could distinguish “this firm talks this way” from “this firm started talking this way” — with five years of panel, that is a real cost. Second, and more serious: what the frame-level model left unexplained was, by construction, whatever is persistently different about a firm relative to the corpus, and that residual firm effect is not automatically washing. It can just as easily be corporate style, IR culture, sector norms, or technological maturity. The clearest symptom was in the model’s own robustness check: estimating dispersion at the firm level, instead of the document level, made the flagged tail disappear entirely, because all of the between-firm heterogeneity the test was supposed to isolate got absorbed into the dispersion term instead. Adding sector fixed effects to the frame-level model also moved firms in and out of the tail, which is the same problem from a different angle: part of what looked like “excess” was sector composition.

The index used here is estimated at the **firm-year** level instead, which fixes both problems directly: it is a weighted least-squares regression (weights = a firm-year’s frame count, which downweights noisy low-volume years without a separate dispersion parameter) of a firm-year’s promotional rate on its own disclosed-behaviour rate, its activity grounding, its log AI-disclosure volume, its document-form mix, and sector and year fixed effects, with firm-clustered standard errors:

$$\text{PromoRate}_{it} = \alpha + \beta_1 \text{Behavior}_{it} + \beta_2 \text{Grounding}_{it} + \beta_3 \log(\text{Volume})_{it} + \gamma \text{FormMix}_{it} + \text{Sector FE} + \text{Year FE} + \varepsilon_{it}$$

$$\text{Washing}_{it} = \hat{\varepsilon}_{it}$$

Behavior_{it} is the same non-promotional-frame behaviour rate as before, now computed per firm-year. Grounding_{it} is the concreteness of that firm-year’s disclosed-activity inventory (named function, deployed/scaled stage, named product, quantified outcome, named provider), shrunk with an empirical-Bayes prior toward the corpus mean, because 51% of firm-years report zero disclosed activities that year and a raw share would collapse to zero for half the panel from missing data, not from an absence of substance. Sector and year absorb exactly the composition effects that the old design’s robustness checks showed were contaminating it. Washing_{it} is the residual, standardized against its implied binomial variance into a z-score: continuous, signed, and comparable across firm-years.

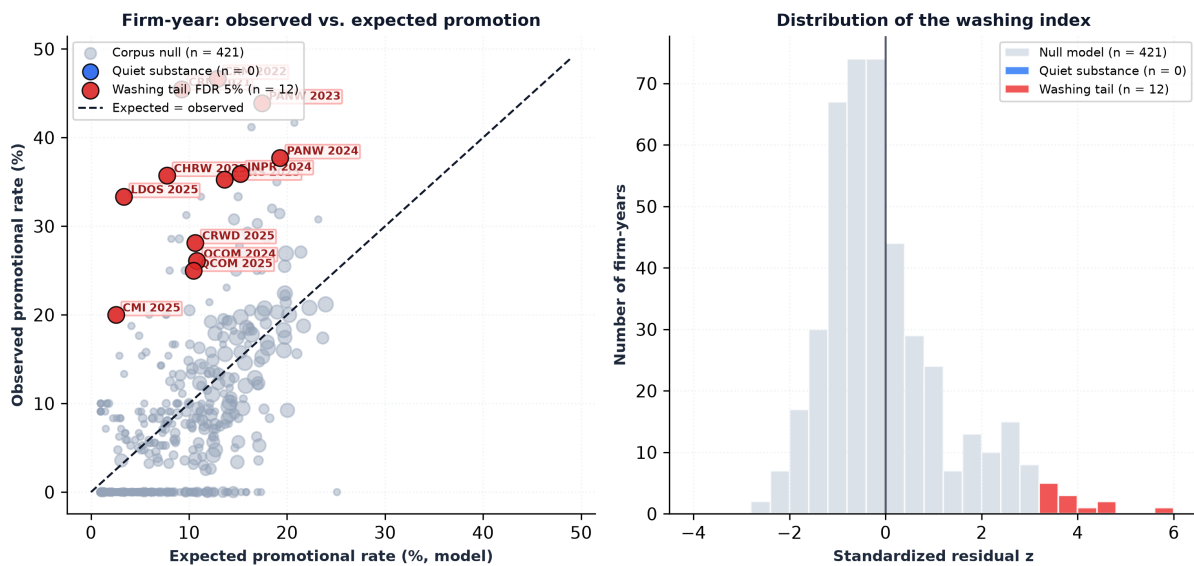


Figure 9: The AI-washing index (firm-year panel): observed vs. expected promotional rate, with the 12 firm-years in the washing tail at 5% FDR annotated (left); distribution of the standardized residual z across all 433 scored firm-years (right).

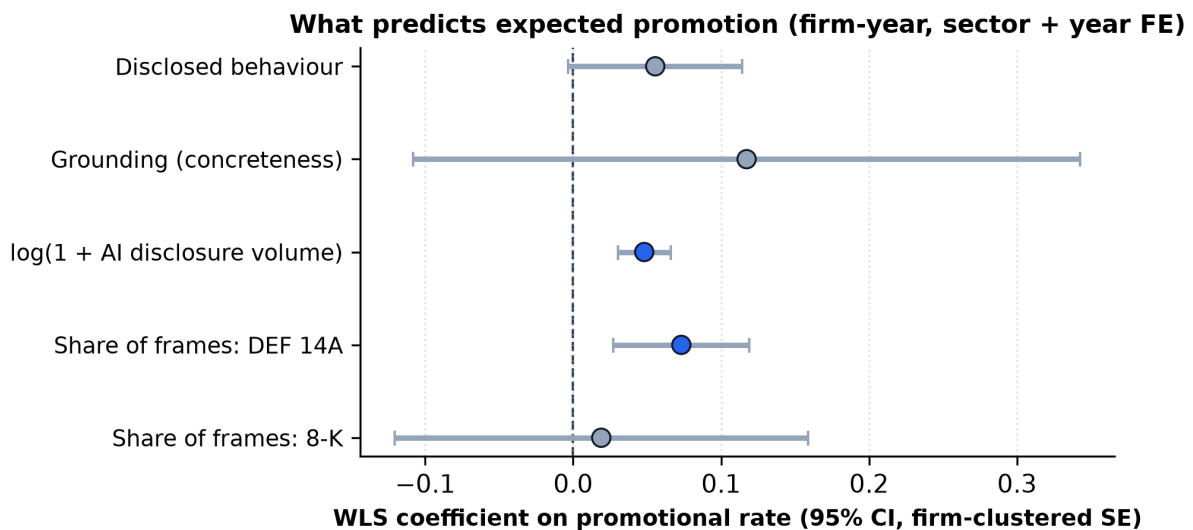


Figure 10: Coefficients of the firm-year expected-promotion model (95% CI, firm-clustered SE, n = 433 firm-years).

Volume and document-form mix (DEF 14A vs. 10-K) are the strongest, most reliably positive predictors of expected promotion; disclosed behaviour is positive and borderline significant ($\beta = +0.056$, $p = 0.064$); grounding is positive but not significant at the firm-year level ($\beta = +0.117$, $p = 0.308$) once sector, year, and the other covariates are already in the model, though it is significant in the pooled cross-section used below ($\beta = +0.198$, $p = 0.042$). Only what exceeds this firm-year expectation counts as washing.

A Benjamini-Hochberg cut at 5% FDR over 433 firm-years with at least 10 frames flags **12 firm-years** promoting more than expected — LDOS 2025, CDNS 2025, PANW 2023, CRM 2021, CHRW 2025, CRM 2022, PANW 2024, JNPR 2024, CMI 2025, QCOM 2025, QCOM 2024, CRWD 2025 — and **none**

promoting less; three firms (CRM, PANW, QCOM) are flagged in more than one year. This is a secondary, binary summary of the continuous index, not the primary result.

Persistence. The firm-year design allows a check the pooled version could not: does a firm's washing residual in year t predict its residual in year $t+1$? Spearman correlation across 198 consecutive firm-year pairs is $+0.379$ ($p < 0.001$) — clearly positive, well short of 1. That is close to the ideal answer for a construct that is meant to capture something real but not perfectly sticky: a pure firm fixed effect (corporate style that never changes) would show much higher persistence, and pure noise would show none.

Sensitivity to specification. A cross-sectional (pooled 2021–2025) version of the same regression, used for the descriptive cross-firm comparisons below and in Appendix C, flags 13 of 306 firms as a washing tail. Across three variants (with/without sector fixed effects, unique vs. non-deduplicated frames), 9 firms are flagged under every specification tested: **CDNS, CMI, CRWD, HUM, JCI, MS, NOC, PANW, QCOM**. Dropping sector fixed effects widens the tail to 16 (Jaccard 0.61 against the reference), confirming again that sector composition is part of what a less-controlled specification would misread as washing.

Does the index carry economic information? Chapter 5 answers this by adding the firm-year washing index as a regressor on top of AI-disclosure volume, semantic content, and disclosed activity ($Y = \text{Volume} + \text{Content} + \text{Behaviour} + \text{Washing} + \text{Controls} + \text{FE}$). Because the index now genuinely varies within firm across years, this is a stronger test than the pooled version could support: the washing index adds essentially nothing for any of the five outcomes ($\Delta R^2 \leq +0.002$, $p \geq 0.145$ in every case, including R&D-to-sales, $\Delta R^2 = +0.0006$, $p = 0.447$). Concrete disclosed behaviour and specificity carry economic information; a firm-year's promotional excess, once behaviour and grounding are already controlled for, does not.

What the index does not establish. The one case with independent, external evidence of a washing concern, Welltower's 2025 SEC comment letter, sits at the 90th percentile of the pooled cross-section ($z = +2.02$) but is not flagged after FDR correction across 306 firms; its own 2025 firm-year, the year closest to the letter, is unremarkable ($z = +0.67$). The three lexical false positives from the same search (ANET, HPE, NVDA — comment letters about segment accounting and revenue recognition, not AI disclosure) rank well below Welltower, at the 30th, 2nd, and 25th percentiles respectively, so the index at least discriminates the real case from the noise cases even where it doesn't cross the formal threshold. This is a documented limitation, not evidence against the index: the SEC's concern in the Welltower matter was that earnings-call language was unsupported by proportional 10-K disclosure, a cross-venue phenomenon a within-filing index cannot see by construction, since it never compares a firm's call language to its own filing language. That comparison is a separate, complementary result reported in Chapter 6. Full specification-sensitivity, placebo, and split-half results are reported in Appendix C.

4.5 ECONOMIC DISTINCTIVENESS

Unlike the activity cross-tabulation above, the financial and market variables used here never passed through the LLM extraction pipeline at all; they come from XBRL filings and market prices, an entirely separate data source from the text the archetypes are built on. This is genuine external evidence, but it answers a narrower question than “are the archetypes real”: it asks whether the archetypes correspond to economically distinct *firms*, not whether the posture dimensions themselves are valid. An archetype with no economic footprint would still be a real, reproducible communication strategy (Table “Choosing k by bootstrap stability” already establishes that on its own terms); it would just mean that strategy is not one that shows up in fundamentals, which is itself an informative finding, not a failure of the clustering.

The economic-distinctiveness analysis pools all firm-years with filings from 2021 to 2025 (2,475 firm-years, 508 firms; archetype counts: 1,425 No AI, 450 Vocal Substantives, 350 Defensive Disclosers, 250 Governance-Led Disclosers) and tests whether the archetypes correspond to economically distinct firms, both crudely and after sector-by-year adjustment ($X = \alpha[\text{sector} \times \text{year}] + \gamma \cdot \text{Archetype} + \epsilon$, referenced against No AI, firm-clustered standard errors).

Crude medians already put Vocal Substantives ahead of the other two archetypes on the dimensions that matter most for a technology-intensive segment: R&D/sales 12.4% vs. 5.7–6.2% for the other archetypes, gross margin 61.1% vs. 41.5–44.8%, beta 1.09 vs. 0.79–0.81. The residualized panel, within the same sector and year, confirms each archetype separately:

- **Vocal Substantives** is the most consistently distinct archetype: +0.44 log points of market cap ($p < 0.001$), +5.6 percentage points of R&D-to-sales ($p < 0.001$), +11.0 points of gross margin ($p < 0.001$), +0.10 of beta ($p = 0.008$), and +0.97 of price-to-sales ($p = 0.028$), significant on five of the nine dimensions tested.
- **Governance-Led Disclosers** separate from No AI mainly on scale and earnings quality rather than technology intensity: a market-cap premium (+0.33 log points, $p = 0.002$) and a ROIC-minus-WACC premium (+0.065, $p = 0.007$), but *lower* price-to-sales (−0.86, $p = 0.023$) and no R&D or margin premium. This is consistent with the archetype being defined by governance-heavy disclosure that correlates with firm size and regulatory exposure (larger, more heavily regulated firms disclose more governance content) rather than with distinctive AI economics.
- **Defensive Disclosers** barely separate from No AI at all: a small R&D premium (+3.6 points, $p = 0.010$) is the only dimension reaching significance out of nine tested.

No archetype shows a robust advantage in every dimension tested. The story that survives sector-year adjustment is that the archetype most concretely grounded in disclosed conduct (Vocal Substantives, [Table 15](#)) is also the one most consistently distinct on firm fundamentals, not that talking about AI, or emphasizing governance, is by itself.

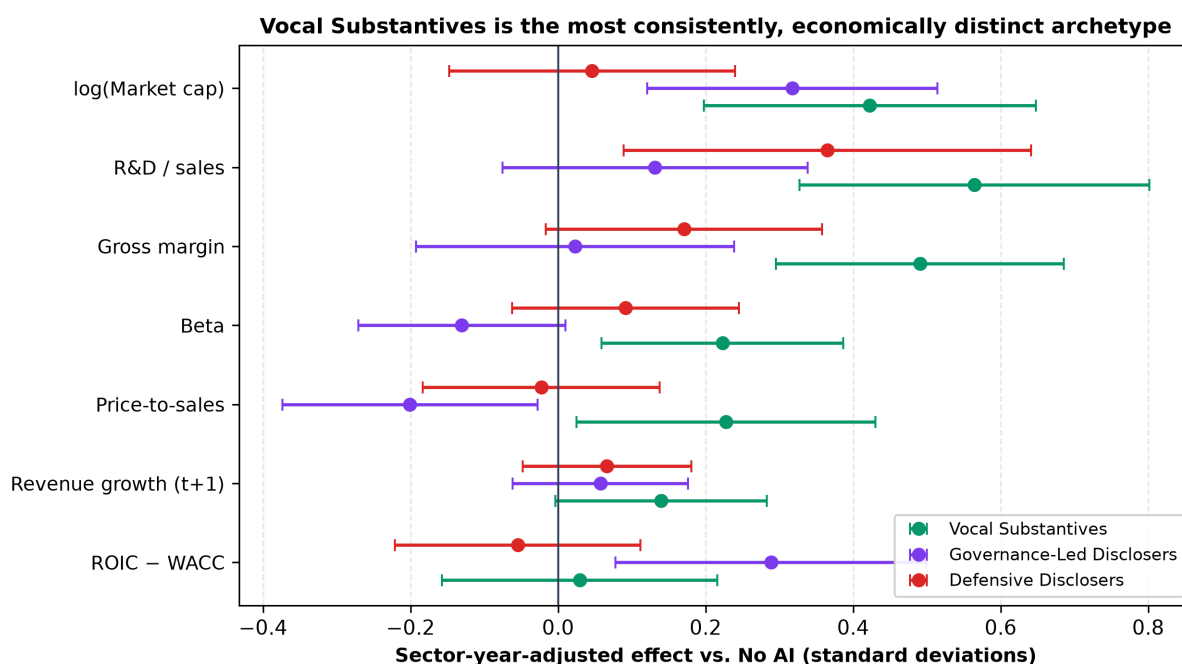


Figure 11: Sector-year-adjusted effect of disclosure archetype on firm fundamentals, relative to No AI (95% CI, firm-clustered SE). Vocal Substantives separates most consistently from the sector-year baseline.

5. DOES AI DISCLOSURE CONTENT CARRY ECONOMIC INFORMATION?

5.1 DOES DISCLOSURE CONTENT ADD ECONOMIC INFORMATION?

The central empirical test of the thesis asks a simple question in a careful way: once you already know how much a firm discloses about AI, and you already know its financial fundamentals, does *what* it says add anything? Five financial outcomes are tested (beta, idiosyncratic volatility, price-to-sales, R&D-to-sales, and next-year revenue growth) starting from the full 2021–2025 panel (2,475 firm-years, 508 firms). Each outcome then drops firm-years missing that outcome or a control variable (R&D/sales, for instance, has much thinner XBRL coverage than market cap) and firm-years whose sector-by-year cell has fewer than two firms, since a fixed effect needs at least two observations to be estimated; the resulting samples, reported in [Table 17](#), range from 649 firm-years (R&D-to-sales) to 1,039 (price-to-sales).

For each outcome, four models are built on top of each other:

- **M0**: a baseline with just financial fundamentals.
- **M1**: the baseline plus how much a firm discloses about AI.
- **M2**: M1 plus seven dimensions of what it discloses (how realized, how deployment-focused, how risk-focused, and so on).
- **M3**: M1 plus which disclosure archetype the firm belongs to, as an alternative to the content dimensions.

The number that matters is how much better M2 predicts the outcome than M1 alone: that gap is the information content adds once volume is already accounted for. That gap is checked three different ways so a single lucky specification can't drive the result: a bootstrap confidence interval, a statistical test on the seven content coefficients, and a permutation test that shuffles the content variables across firms to see what a coincidence would look like.

The content block adds a partial R^2 of 1.8–6.0% across all five outcomes:

- **Beta:** +3.3 points (bootstrap CI [+1.5, +9.7], Wald $F = 3.09$, $p = 0.004$, permutation $p = 0.005$).
- **Idiosyncratic volatility:** +1.8 points (CI [+1.0, +5.7], Wald $p = 0.203$, permutation $p = 0.050$).
- **Price-to-sales:** +2.0 points (CI [+1.4, +8.3], Wald $p = 0.048$, permutation $p = 0.020$).
- **R&D-to-sales:** +6.0 points (CI [+3.4, +15.8], Wald $p = 0.001$, permutation $p = 0.005$).
- **Next-year revenue growth:** +5.2 points (CI [+0.9, +13.2], Wald $p < 0.001$, permutation $p = 0.005$).

Every bootstrap interval excludes zero, adjusted R^2 rises with the content block in all five outcomes, the Wald test rejects at 5% in three of five (idiosyncratic volatility falls short at $p = 0.203$), and the sector-by-year permutation, which destroys only the content-outcome association while preserving sector, year, fundamentals, and volume, rejects in all five. The magnitude is modest but real: of the same order as what disclosure volume itself contributes (0.2–5.6 points) and larger than what segment membership contributes as a dummy (0.1–1.7 points).

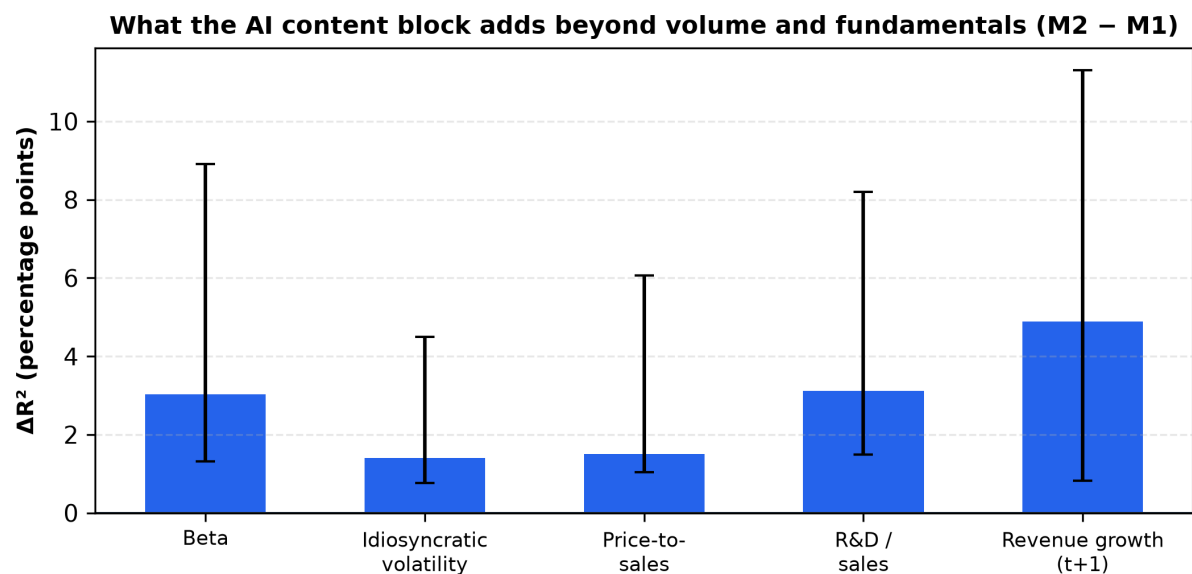


Figure 12: Incremental explanatory power of AI disclosure content beyond volume and fundamentals: partial R^2 by outcome, with bootstrap 95% CI (firm-clustered).

Standardized coefficients in M2 identify **specificity** (how much of a firm's AI language comes with named processes, products, providers, figures, or dates) as the dimension carrying the signal in four of five outcomes ($\beta = +0.37$ for beta, $+0.32$ for idiosyncratic volatility, $+0.45$ for price-to-sales, $+0.37$ for revenue growth, all $p < 0.01$). Promotional language contributes essentially nothing ($\beta \approx -0.1$ in price-to-sales, $p = 0.07$; indistinguishable from zero elsewhere), and governance content is associated with lower beta (-0.12 , $p = 0.04$).

Given volume, the firm that discusses AI concretely carries more idiosyncratic risk, trades at a higher multiple, and grows faster than the firm that discusses AI in the abstract; the firm that merely promotes does not, consistent with verifiable detail serving as a costly, credible signal rather than uninformative cheap talk (Spence 1973; Rogers, Van Buskirk, and Zechman 2011).

5.2 ROBUSTNESS CHECK: MIX OR VOLUME?

Re-expressing the content block as within-firm semantic shares (each concept's frames as a fraction of the firm's total AI frames, with volume entered separately) rather than log-intensities halves the ΔR^2 in idiosyncratic volatility, price-to-sales, and growth and the Wald test stops rejecting in those three outcomes, while beta and R&D-to-sales are essentially unaffected. Part of the content signal is therefore the intensity of each dimension and not only the mix; the compositional signal alone survives only where systematic risk and R&D are concerned.

5.3 WHAT VS. HOW

Six activity-count families (customer-facing deployment, internal deployment, own AI, named provider, infrastructure, quantified outcome, log-transformed intensities with zeros) are entered as an alternative content block and as an additional block layered on top of the semantic dimensions. The two blocks are not redundant: together they raise ΔR^2 to 2.1–7.8 points across the five outcomes, and each block retains at least half its individual contribution once the other is already included, with two informative exceptions:

- **Idiosyncratic volatility:** activity absorbs most of the semantic signal (semantics given activity $\Delta R^2 = +0.009$, $p = 0.50$). Idiosyncratic risk tracks what a firm does more than how it narrates it.
- **Revenue growth:** activity contributes roughly 2.4 times what semantic style contributes once each is conditioned on the other (+0.029 given semantics vs. +0.012 given activity). Growth is anticipated more by the concrete activity a firm describes than by its rhetorical specificity, though both remain jointly informative.
- **Beta and price-to-sales:** both blocks hold up independently, each still significant conditional on the other.

The joint reading: specificity of language matters in addition to disclosed activity, not as a substitute for it, and where the outcome is fundamentally about “doing,” such as idiosyncratic risk and growth, the activity block dominates.

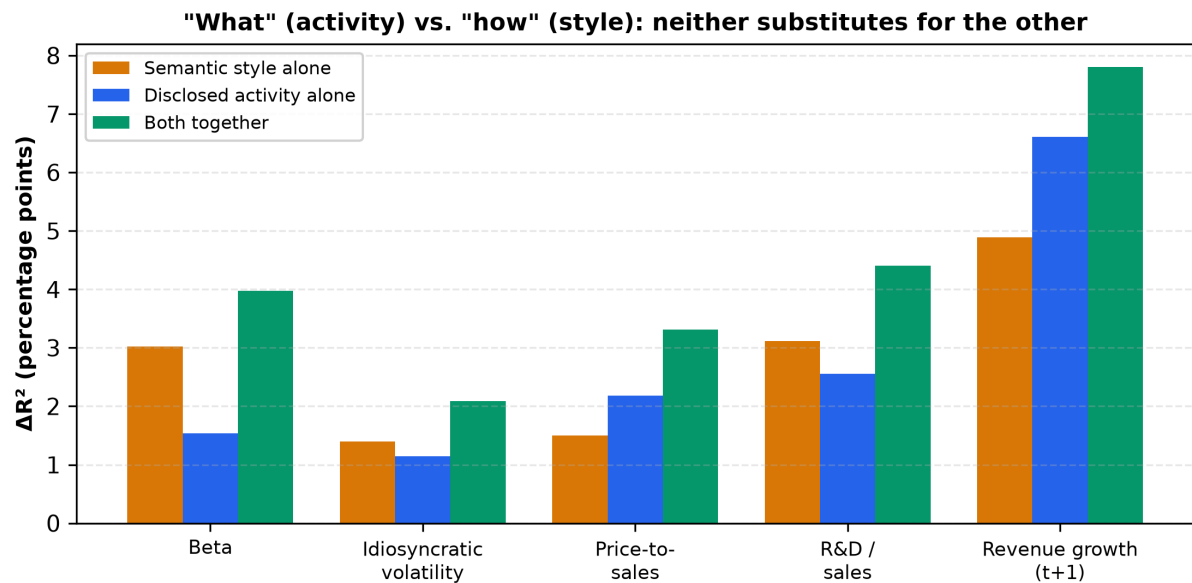


Figure 13: What vs. how: ΔR^2 contributed by semantic style alone, disclosed activity alone, and both together, by outcome.

[Table 17](#) reports the central empirical test of whether textual AI disclosure carries incremental economic information beyond disclosure volume and financial fundamentals. Across five financial outcomes, Model 1 (M1) includes firm size, sector-by-year fixed effects, disclosure volume, and standard financial controls. Model 2 (M2) adds the seven semantic frame dimensions (not disclosed activity, added separately as its own block below). In every case, semantic content adds statistically significant explanatory power (partial R^2 of 1.8 to 6.0 percentage points, bootstrap intervals excluding zero), rising to 2.1–7.8 points once disclosed activity is layered on top of the semantic block.

Table 17: Central incremental-information results.

Outcome	n	R2 M1	R2 M2	DeltaR2 content	95% bootstrap CI	Partial R2	Wald p	Permutation p
Beta	1,037	0.095	0.125	0.03	[+0.013, +0.089]	3.3%	0.004	0.005
Idiosyncratic volatility	1,037	0.211	0.225	0.014	[+0.008, +0.045]	1.8%	0.203	0.05
Price-to-sales	1,039	0.235	0.25	0.015	[+0.010, +0.061]	2.0%	0.048	0.02
R&D/sales	649	0.479	0.51	0.031	[+0.015, +0.082]	6.0%	0.001	0.005
Revenue growth (t+1)	1,003	0.067	0.116	0.049	[+0.008, +0.113]	5.2%	<0.001	0.005

5.4 DOES PROMOTIONAL EXCESS ADD ANYTHING ON TOP?

The cleanest version of the question this chapter asks is:

$$Y_{it} = \alpha + \beta_1 \text{Volume}_{it} + \beta_2 \text{Content}_{it} + \beta_3 \text{Behaviour}_{it} + \beta_4 \text{Washing}_{it} + \text{Controls} + \text{FE} + \epsilon_{it}$$

where *Washing* is the continuous AI-washing index built in Chapter 4: a firm-year residual (promotional rate observed minus expected, given that firm-year's own disclosed behaviour, grounding, volume, and document mix, with sector and year fixed effects), estimated on the firm-year panel — not a pooled firm effect frozen across years, so it carries real within-firm variation over time. Adding it as a fourth block on top of volume, semantic content, and disclosed activity asks whether a firm-year's *residual* promotional tone, the part not already explained by what it discloses doing and how grounded that is, carries any further information about firm fundamentals.

The outcome closest to a firm's actual technological intensity, R&D-to-sales, anchors the reading: adding Washing raises R^2 from 0.5227 to 0.5232, a ΔR^2 of +0.0006 that is not distinguishable from zero (Wald $F = 0.58$, $p = 0.447$). The same null holds for beta ($\Delta R^2 = +0.0020$, $p = 0.208$), idiosyncratic volatility ($\Delta R^2 = +0.0017$, $p = 0.145$), price-to-sales ($\Delta R^2 = +0.0014$, $p = 0.224$), and revenue growth ($\Delta R^2 = +0.0002$, $p = 0.615$) — the largest of the five ΔR^2 is 0.002 percentage points, an order of magnitude below what volume, content, or activity each contribute on their own.

Read against [Table 17](#) and the “what vs. how” decomposition above, the pattern across all four blocks is consistent: volume, specificity, and disclosed activity each add real, robust explanatory power; a firm's promotional excess, once behaviour and grounding are already in the model, adds essentially none. Concrete disclosed behaviour and specificity carry economic information; promotional excess does not.

6. SAME FIRM ACROSS VENUES AND OVER TIME

6.1 CALLS VS. FILINGS

The most tightly identified result in the thesis compares the same firm, in the same fiscal year, across the two venues that differ most in liability exposure: the earnings call, an informal and voluntary disclosure venue (Frankel, Johnson, and Skinner 1999; Matsumoto, Pronk, and Roelofsen 2011), and the SEC filing, a monitored venue carrying legal liability for material misstatement.

The design uses firm $\times$ fiscal-year fixed effects with exactly two channel observations per cell, so the estimator reduces to a within-cell difference: call intensity minus filing intensity. This cancels everything common to the firm in that period, including the industry-wide AI boom itself. Because venue bundles legal exposure with differences in audience, format, length, and topic selection, the estimate is interpreted as a within-firm, within-year venue-associated difference, not as the isolated causal effect of legal liability.

The extensive-margin sample comprises 2,281 firm $\times$ fiscal-year cells across 479 firms (449 present on both sides of the 2024 cutoff), aligned by the fiscal period each document actually covers rather than its filing date. Of these cells, 1,021 contain calls that mention no AI at all (5,238 of 7,947 transcripts).

The denominator matters here: a scorable call paragraph runs to 129 words on average against 67 for a filing paragraph, so a raw per-paragraph count would overstate the gap by roughly half. This thesis reports the rate per 1,000 words instead. On that basis, the same firm, in the same fiscal year, places 9.9 $\times$ as many AI-promotional statements in its earnings call as in its filing, and 17.1 $\times$ as many quantified claims, even though the filing has a higher share of documents mentioning AI at all (65% vs. 55%). Chapter 3 already showed that what firms added to their filings between 2023 and 2025 was, above all, an AI risk factor and a governance line: this cross-venue result is consistent with that pattern, and suggests the product pitch is being made somewhere else.

[Table 18](#) and [Figure 14](#) report the same comparison two ways: the table gives the channel means and the call-to-filing ratio, and the figure gives the within-cell difference with a firm-clustered 95% bootstrap confidence interval.

Table 18: Channel means and call-to-filing ratio, per 1,000 words.

Measure (per 1,000 words)	Filing	Call	Call / Filing
Promotional claims	0.006	0.054	9.9 $\times$
Quantified claims	0.003	0.052	17.1 $\times$

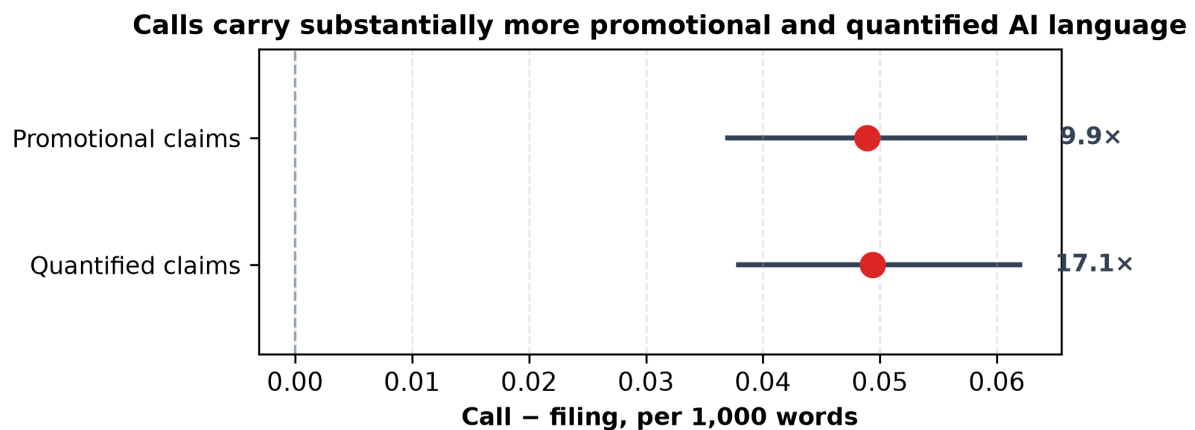


Figure 14: Within-firm, within-year call-minus-filing difference in AI disclosure rate, with a firm-clustered 95% bootstrap confidence interval. Ratios (call / filing) are annotated on the right.

Section 6.2 provides an independent, denominator-free check by comparing the same firm's *disclosed activities* rather than word or paragraph counts across venues, and finds the activity mix itself essentially unchanged while what is attached to it (names, figures, providers) differs sharply, the most robust reading of this gap. This is a promotional gap between venues, not evidence that any individual claim is false, but it is precisely the type of cross-venue mismatch the SEC flagged in the Welltower matter (Strauss et al. 2025; Bozanic, Dietrich, and Johnson 2017), measured here across the entire corpus.

6.2 ACTIVITIES ACROSS VENUES

Comparing disclosed-activity composition across 459 firm $\times$ fiscal-year cells (183 firms with at least three activities in each channel), the *mix* of what firms say they do is essentially identical across venues: customer-facing deployment, internal deployment, and own-AI shares differ by only 0.6 to 2.2 percentage points, none distinguishable from zero at 5%.

What differs is the evidence attached to the same claim. The call adds:

- a declared business function (+11.7 points);
- a quantified outcome (+11.0 points; the filing almost never states the figure);
- a named external provider or partner (+5.3 points, nearly triple); and
- a named product (+3.8 points),

while the filing retains relatively more infrastructure investment, talent, and governance language. The gap is not “different activities depending on the audience” but “the same activity, with figures and names for the analyst and without them in what is signed,” a pattern stable across fiscal years, with the quantified-outcome gap ranging from +9.0 to +11.8 points between 2021 and 2025.

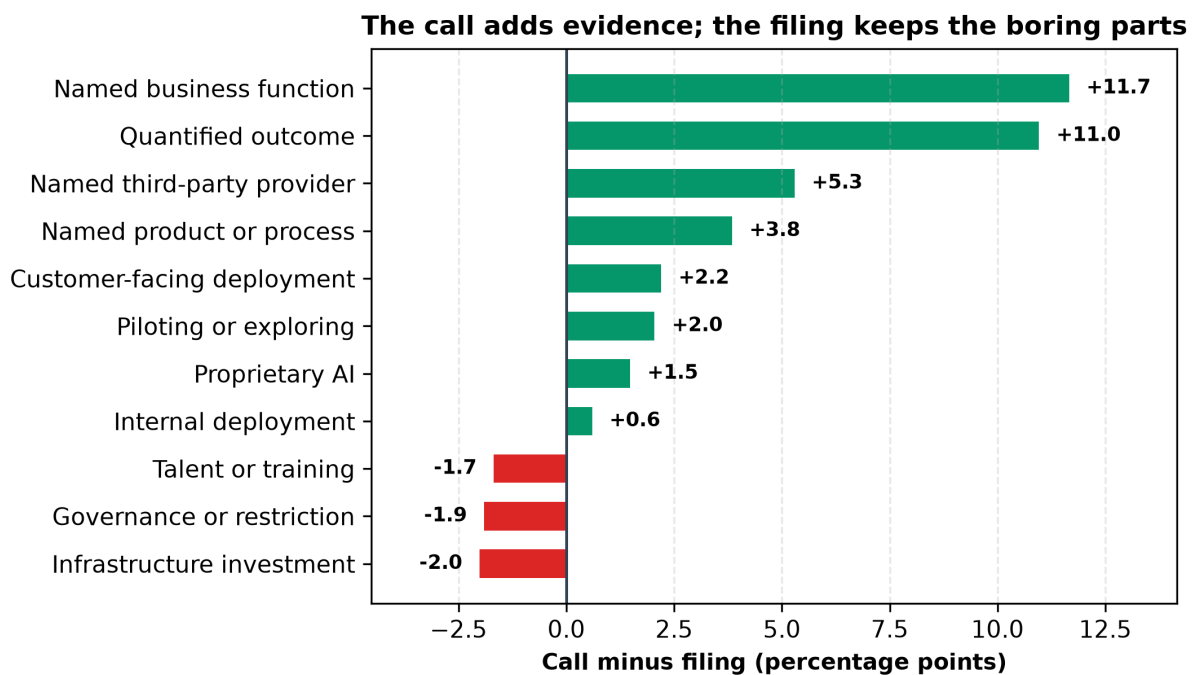


Figure 15: Same activity, different evidence: call-minus-filing gap in how often each activity carries a given piece of evidence, ordered from largest to smallest (percentage points).

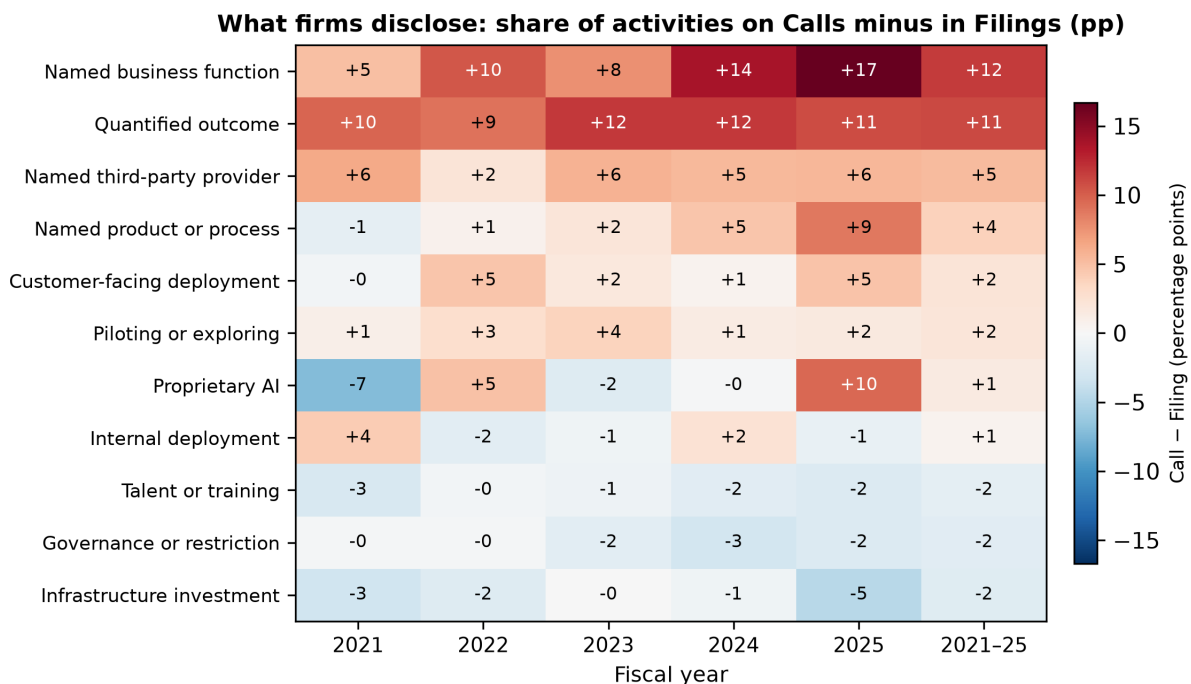


Figure 16: Cross-venue activity gap: comparison of disclosed AI activity frequencies, concreteness, and evidence grounding between formal SEC filings (10-K/10-Q) and earnings conference calls.

6.3 VENUE GAP OVER TIME

With firm fixed effects on the gap itself, the promotional gap (per 1,000 words) widens from a flat base of 0.019 in 2021–2022 to 0.054 in fiscal 2023, 0.077 in 2024, and 0.080 in 2025 (post-2024 coefficient +0.047, $p < 0.001$, on the same 449 firms present on both sides of the cutoff). Governance is the one dimension where the gap does not grow: filings add governance content at roughly the same rate calls add promotion, so the difference nets to zero.

Pre-trend tests fail everywhere, and they have to: the gap was already widening in fiscal 2023, a full year before the SEC’s public warning. So the SEC’s 2024 scrutiny is not separably identifiable against this gap. A rate-normalized check (conditioning on cells where both channels discuss AI, so the comparison is not confounded by the extensive margin) confirms the same null: the post-2024 coefficient on the conditional promotional rate is +0.015 ($p = 0.36$), with flat pre-trends.

Firm-level trajectories vary in level but not in this general pattern:

- NVDA’s gap widens from +0.29 to +0.44 since 2023.
- PLTR stays persistently above +0.3.
- HPE is a rare case where the gap closes, from +0.25 to –0.03.
- Welltower itself, the case motivating the design, has too little AI-call content before 2025 to estimate a firm-specific trajectory.

The firm-level gap does not positively correlate with the AI-washing index of Chapter 4 (Spearman -0.28 , $n = 167$); if anything, firms with a larger cross-venue gap show slightly *lower* within-filing promotional excess, not higher — a reminder that these are different forms of divergence, measured within a filing versus across venues, not two readings of the same underlying quantity.

6.4 SEC 2024

To test this, every firm-quarter (11,302 of them, across all 510 firms) is scored on how much AI exposure it had before 2023, and the design asks whether firms with more pre-existing exposure changed differently around the SEC’s warning than firms with less.

The standard check for this kind of design is whether the two groups were already on different trends *before* the event: if they were, any “effect” you find afterward could just be that pre-existing trend continuing, not something the SEC caused. Here, that check fails for every outcome tested (promotional language, risk, hypothetical framing, total AI volume, specificity) because more-exposed firms were already pulling away from less-exposed firms well before December 2023. The design cannot separate a scrutiny effect from a trend that was already there.

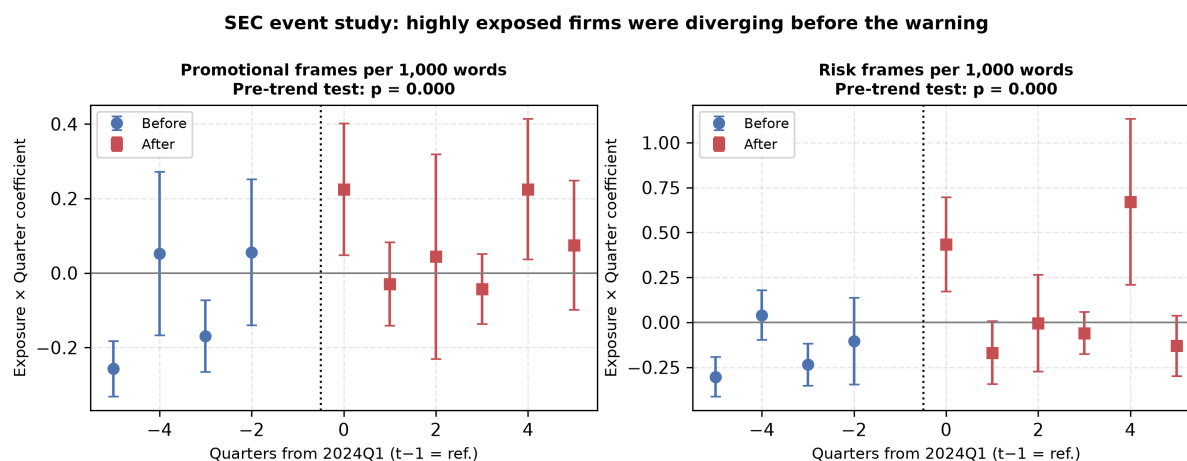


Figure 17: Diagnostic event study: SEC regulatory sweep and enforcement inquiries around AI disclosures across event quarters, showing pre-trend divergence.

A simpler difference-in-differences design (108 more-promotional firms vs. 382 controls, split by pre-period promotional intensity) finds that the more-promotional group added more governance and specificity after December 2023 than the control group (+0.0084 and +0.0161 per 1,000 words, $p = 0.001$ and $p < 0.001$), consistent in direction with a scrutiny response; pre-trends for both outcomes are clean ($p = 0.138$ and $p = 0.245$), though the same-dimension reference outcome (promotional intensity itself) has a pre-trend that sits at the margin of significance ($p = 0.053$), and the jump concentrates in the 2024 filing season rather than tracking the enforcement timeline precisely.

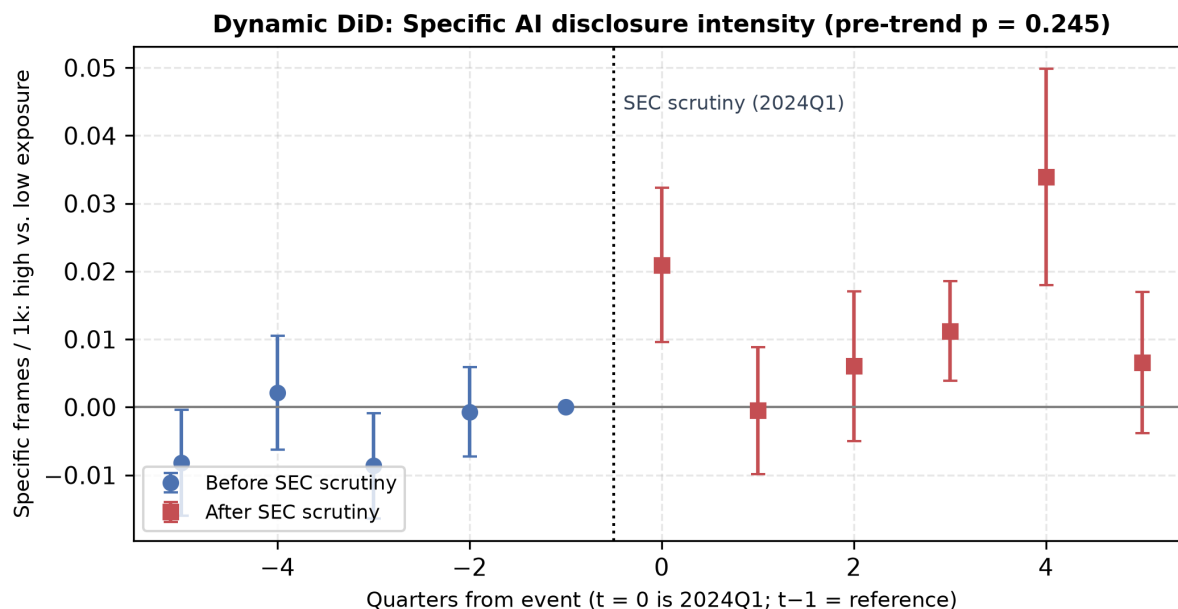


Figure 18: Difference-in-differences dynamic trajectory: specific disclosure intensity per 1,000 words following SEC regulatory scrutiny.

Identification verdict: the causal effect of SEC scrutiny cannot be separated from the concurrent generative-AI boom with this design. What is observed after 2024 (more risk disclosure, more governance, a shrinking promotional share in Chapter 3) is equally consistent with a scrutiny response and with what any firm adds to its 10-K once a technology becomes material.

One sentence on the second shock examined: DeepSeek was also tested (a January 2025 cutoff, the same event-study design, 5,287 firm-quarters), but no discontinuity separable from prior trends was identified; the full analysis, including the failed pre-trend tests, is reported in Appendix E.

Firm-level trajectories and event-study specifications across alternative outcomes are detailed in Appendix E.

Table 19: SEC identification verdict.

Design	Pre-trend test	Verdict
Event study, continuous exposure, firm + quarter FE	Fails for all 5 outcomes ($p < 0.001$)	Not identifiable
Simple DiD, promotional split	At the margin ($p = 0.06\text{--}0.09$); jump concentrated in 2024 10-K season	Not identifiable as a scrutiny effect
Within-firm channel gap (call – filing), post-2024	Fails ($p < 0.01$); gap was already widening in FY2023	Gap persists; no closure after scrutiny
DeepSeek event study (Appendix E)	Fails for 4 of 5 outcomes	Not identifiable

7. AI WASHING AND MANAGERIAL IMPLICATIONS

This chapter draws out the implications of the AI-washing index built in Chapter 4: a firm-year residual — promotional rate observed minus expected, given that firm-year’s own disclosed behaviour, its grounding (concreteness), its overall AI-disclosure volume, and its document-type mix, within sector and year. Twelve firm-years (three of them repeat appearances by CRM, PANW, and QCOM) promote more than the index predicts at 5% FDR; a cross-sectional version of the same regression flags 13 of 306 firms pooled over 2021–2025, of which 9 (CDNS, CMI, CRWD, HUM, JCI, MS, NOC, PANW, QCOM) stay in the tail under every specification tested. Chapter 5 already showed the index adds essentially no further economic information beyond disclosed behaviour and content ($\Delta R^2 \leq +0.002$, $p \geq 0.145$ on all five outcomes). What follows asks what the index is good for instead: reading behavioural grounding, and drawing issuer- and investor-facing implications.

7.1 BEHAVIOURAL GROUNDING

Across the corpus, promotion with concrete activity behind it and promotion without concrete activity behind it are both common, and neither implies the other: 78 of the 90 firms in the top quintile of promotional rate have at least one deployed activity with a named product (median 103 activities), while the remaining 12 (JPM, GILD, CMI, IVZ, GL, PVH, PCAR among them) have a median of five. Promotability, in other words, is not by itself an absence of substance; grounding is the additional evidence needed to tell the two populations apart, and it is the same variable that separates the Vocal Substantives archetype from the promotion-without-support firms inside Chapter 4’s Defensive and Governance-Led Discloser archetypes.

7.2 ISSUER IMPLICATIONS

The measurement built across this thesis converts directly into a disclosure checklist. Credible AI disclosure:

- names a concrete activity, not “AI capabilities,” which 86% of firms use as an empty placeholder object;
- states a clear stage (deployed or scaled, not merely explored or piloted);
- names a specific product or process;

- reports a quantified outcome;
- discloses the source of the AI, including provider and partner relationships, since 16% of activities in the corpus leave this unstated, and undeclared origin is far more common among Defensive Disclosers and Governance-Led Disclosers (33% each) than among Vocal Substantives (10%); and
- is consistent between what is said on the earnings call and what is filed with the SEC, the dimension issuers have the least visibility into internally, since the SEC's own enforcement rationale in the Welltower matter targets exactly this consistency, not the promotional content of the filing per se.

7.3 INVESTOR IMPLICATIONS

Of the textual dimensions available in AI disclosure, specificity carries incremental economic information (Chapter 5: 1.8–6.0 points of partial R^2 in beta, idiosyncratic volatility, price-to-sales, R&D-to-sales, and next-year revenue growth), while promotional tone carries essentially none. Disclosed activity adds information beyond semantic style, and dominates it specifically where the outcome concerns operating risk and growth.

Investors should read earnings-call AI claims against what the same firm has filed: the same activity mix is described in both venues, and the call does supply concrete evidence the filing typically omits (quantified outcomes, named providers, named products) alongside more promotional language, so cross-venue consistency, not the call's promotional tone by itself, is the more useful credibility signal. This is a gap that has been widening since the 2023 AI boom and has shown no sign of narrowing since the SEC's December 2023 warning.

7.4 DISCUSSION AGAINST THEORY

Three theories fit different parts of the results:

1. The specificity result (Chapter 5) fits a signalling account (Spence 1973; Rogers, Van Buskirk, and Zechman 2011): specific, falsifiable claims (named products, quantified outcomes) carry information that generic promotion does not. This is consistent with investors pricing that distinction, but does not prove it: the underlying regressions are cross-sectional associations, not an event-based test of market reaction.
2. The near-disappearance of the “no AI” segment by 2025, together with Governance-Led Disclosers and Defensive Disclosers showing no R&D or margin premium over non-disclosing peers in their own sector and year even though Governance-Led Disclosers carries a market-cap and earnings-quality premium driven by firm size and regulatory exposure rather than technology intensity (Chapter 4), fits a legitimacy account (DiMaggio and Powell 1983; Suchman 1995) better than a genuine-adoption account for roughly two-thirds of the disclosing population: most firms appear to be talking about AI because their peers are, not because it changes their economics.
3. The widening, unclosed gap between calls and filings (Chapter 6) looks more like cheap talk in the unmonitored channel (Crawford and Sobel 1982; Frankel, Johnson, and Skinner 1999) than like costly signalling, and this cross-venue gap is the closest match to what the SEC has actually enforced (Gensler 2023; U.S. Securities and Exchange Commission 2024; Strauss et al. 2025).

The washing index is best read through Meyer and Rowan's (1977) idea of decoupling: formal disclosure running ahead of what a firm actually does (Westphal and Zajac 1994; Bromley and Powell 2012; Crilly, Zollo, and Hansen 2012). It does not observe actual operations directly; it catches the

mismatch between voice and disclosed conduct within a filing, against a firm's own expected rate. The cross-venue gap of Chapter 6 catches a related but distinct mismatch, across venues, and is reported separately rather than folded into the same score. One limitation runs through both: greenwashing research can check claims against a hard number, like emissions data; AI washing as measured here has no equivalent external benchmark (Delmas and Burbano 2011; Lyon and Montgomery 2015; Marquis, Toffel, and Zhou 2016). The index is validated internally instead (against a placebo, a split-half check, agreement across specifications), not against an outside record of what firms actually did. That is stated here as a limitation, not glossed over.

Table 20: Issuer / investor checklist.

Dimension	Issuer guidance	Investor read
Named activity	State the specific action-object, not "AI capabilities"	Absence is the single most common gap (86% of firms)
Stage	Report deployed/scaled, not only piloted	Piloting-only claims should be discounted relative to deployed claims
Product/process named	Name it	Named claims carry more of the incremental economic signal
Quantified outcome	Report the figure in the filing, not only on the call	The filing omits this far more often than the call (61%→37% pooled-to-filing-only)
AI source / provider	Disclose whether AI is own, third-party, or co-developed	16% leave this unstated; undeclared origin clusters in the least-grounded segment
Venue consistency	Match filing content to call content	The 9.9× promotional gap has not closed since 2024 scrutiny

8. CONCLUSIONS

The thesis's main empirical findings are five:

1. AI disclosure grew nearly seven-fold in intensity and reached 92% of S&P 500 firms between 2021 and 2025, with the composition of that growth shifting decisively toward risk and governance content rather than promotion.
2. Disclosure organizes into six conceptually named posture dimensions (promotional, risk, governance, temporal, positioning, and specificity, plus disclosure intensity) deliberately built without any declared-action content, and into a reproducible three-archetype taxonomy over those dimensions (Vocal Substantives, Governance-Led Disclosers, Defensive Disclosers, plus a rule-based No AI segment). A separate disclosed-activity extraction, read cautiously because it shares the same underlying corporate text, characterizes what each archetype reports doing. A genuinely external check against XBRL and market data (never part of the LLM pipeline) shows only Vocal Substantives, the archetype reporting the most disclosed activity even after controlling for total disclosure volume, corresponds to distinct firm economics; the other two archetypes separate mainly on firm size and governance register, not on R&D or margins.
3. Promotional posture does not organize its own archetype at all: it rises with disclosed activity volume ($R^2 = 0.176$, robust to controlling for total disclosure volume), and the residual from that relationship gives a continuous measure of promotional excess in place of an arbitrary tercile cut.
4. The semantic content and disclosed activity of AI disclosure carry modest but robust incremental economic information (1.8–6.0 points of partial R^2 for semantic content alone, 2.1–7.8 points with disclosed activity added) beyond disclosure volume and financial

fundamentals, concentrated in specificity and in concrete activity rather than in promotional tone; anchored on R&D-to-sales as the outcome closest to a firm's actual technological intensity, adding the firm-year AI-washing index on top of volume, content, and activity contributes nothing further ($\Delta R^2 = +0.001$, $p = 0.447$), and the same null holds for beta, idiosyncratic volatility, price-to-sales, and revenue growth ($p \geq 0.145$ in every case): disclosed substance carries information that promotional excess does not.

5. Comparing the same firm in the same fiscal year across venues, the *mix* of disclosed AI activity is nearly identical between earnings calls and SEC filings, but calls attach far more concrete evidence and promotion to that same activity (10 times more promotional statements and 17 times more quantified claims per 1,000 words), a within-firm, within-year venue-associated gap that opened with the 2023 boom and has not closed despite the SEC's 2024 AI-washing enforcement.

AI washing is measured here as a single continuous index estimated at the firm-year level, promotional excess conditional on disclosed behaviour, grounding, volume, document mix, sector, and year; added as a regressor alongside volume, semantic content, and disclosed activity, it adds essentially nothing to fundamentals-adjusted economic outcomes, while concrete, grounded disclosed behaviour and specificity do. The cross-venue gap (Chapter 6) is a related but analytically separate institutional finding, and the closest match to the mechanism regulators have actually enforced so far.

The methodological contribution is a five-layer, largely reproducible extraction pipeline that recovers structured, falsifiable claims and grounded activities from over nine million paragraphs at a cost no purely human-coded design could match, with venue-specific measurement error documented rather than assumed away. The managerial contribution is a concrete disclosure checklist for issuers and a cross-venue credibility framework for investors, both grounded in what the measurement shows carries information.

Theoretically, the results are most consistent with a signalling account of specificity, a legitimacy/institutional-diffusion account of the "no AI" segment's disappearance, and a cheap-talk account of the unmonitored earnings-call channel: three different mechanisms operating simultaneously rather than one master explanation of AI disclosure. The main claims of this thesis are measurement and association rather than causal effects: the most tightly identified design isolates a venue-associated disclosure difference, not the causal effect of legal liability, and the SEC and DeepSeek shock designs fail pre-trend tests and are reported as failed identification rather than assigned a causal reading.

Four limitations are carried through every chapter:

1. Every frame and activity label comes from an LLM judge: the measured reliability (κ 0.81–0.87) is agreement between two LLM judges, not accuracy against an independent ground truth.
2. "Disclosed behaviour" and "disclosed activity" measure what a firm says it does, not what an audited operational record would confirm.
3. The SEC and DeepSeek shocks are not identifiable against the background AI boom with the designs available, a null result reported rather than concealed.
4. Statistical power for firm-level washing claims concentrates in high-disclosure-volume firms, so claims about which firms wash more are properly claims about which firms this corpus can say anything about at all.

Cross-venue results are robust to using words rather than paragraphs as the denominator and to denominator-free activity comparisons.

Future research should extend the cross-venue design to a synthetic-control or matched-donor-pool setting if a denser call-transcript series becomes available, particularly for cases like Welltower with too little pre-period call content to estimate a firm-specific trajectory; and extend the comparative international panels (Chile, Italy) noted in the data section to test whether the venue-gap and segmentation results generalize outside a single, SEC-regulated regime.

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APPENDICES

APPENDIX A: PIPELINE

The full ranking of action × object disclosed corporate activities in the S&P 500 ([Table 21](#) below) complements the top-10 chart already shown in Chapter 3 ("What firms disclose doing").

Table 21: Full action x object ranking, pooled vs. filings-only share of firms, with literal textual examples.

Action x Object family	% Firms (Pooled)	% Firms (Filings only)	Literal textual examples
deploy · AI, unspecified object	88.1%	81.7%	
deploy · product feature or service	60.5%	41.6%	ai engine, interconnect products, advertising tools, ai software
deploy · other	54.8%	32.3%	on-device ai, edge ai, aip, ai networks
deploy · ML or predictive model	49.5%	27.0%	ai models, machine learning models, ai algorithms, machine learning model
deploy · analytics or data platform	48.8%	30.4%	ai platform, ai platforms, expert platform, ai-powered analytics
pilot_or_explore · AI, unspecified object	42.0%	26.1%	
scale · AI, unspecified object	38.6%	26.8%	
deploy · named product or platform	36.9%	23.1%	aiops, aip, embedded ai, mist ai
deploy · automation	36.5%	19.1%	automation, ai and automation, automation tools, automation and ai
invest_infrastructure · AI, unspecified object	34.4%	25.3%	
hire_or_train · talent	31.0%	26.8%	ai talent, talent, employees, workforce
integrate · AI, unspecified object	28.9%	21.7%	
invest_infrastructure · compute and infrastructure	28.0%	17.4%	ai infrastructure, data centers, compute capacity, cloud infrastructure
scale · other	25.7%	10.0%	ai integration, hybrid cloud and ai strategy, edge ai, ai portfolio
scale · product feature or service	25.3%	9.6%	digital capabilities, data ecosystem, efx.ai capabilities, firefly services
measure_outcome · other	23.1%	10.6%	ai orders, ai demand, business performance, inference performance
develop · AI, unspecified object	22.5%	18.9%	
measure_outcome · financial outcome	21.2%	10.4%	ai revenue, productivity, revenue, productivity gains
deploy · compute and infrastructure	21.0%	13.2%	ai infrastructure, ai pc, ai accelerators, ai accelerator
invest_infrastructure · other	20.8%	11.5%	ai research and development, research and development, hybrid cloud and ai strategy, ai research

Notes: Pooled sample covers 510 S&P 500 firms with filings and call transcripts. “Filings only” restricts the denominator to statutory SEC filings (10-K, 10-Q, DEF 14A, 8-K), illustrating the significant contribution of earnings calls to corporate disclosure of commercial tools, partnerships, and quantified metrics.

The full 14-variable disclosed behavioral profile by disclosure archetype is in [Table 22](#) below; the 6 headline variables are shown as [Table 15](#) in Chapter 4.

Table 22: Full disclosed behavioral profile by disclosure archetype (14 variables).

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Disclosed activities per firm (Median)	96	18	8	1
Disclosed activities per firm (Mean)	170.7	23.5	15.1	3.6
Customer-facing activity share (%)	52.1%	23.5%	23.9%	21.7%
Internal process activity share (%)	31.1%	63.6%	62.3%	71.2%
Technology source: Proprietary / Own (%)	72.5%	47.2%	38.9%	33.8%
Technology source: Third-party commercial (%)	9.0%	14.8%	17.7%	14.3%

Behavioral feature	Vocal Substantives (n=124)	Governance-Led Disclosers (n=142)	Defensive Disclosers (n=191)	No AI (n=53)
Technology source: Unspecified provenance (%)	18.5%	38.0%	43.4%	51.9%
Firms naming an own product or brand (%)	98.4%	66.2%	43.5%	15.1%
Firms naming an external provider or partner (%)	89.5%	47.2%	31.4%	15.1%
Firms reporting compute / data center capex (%)	89.5%	51.4%	39.3%	7.5%
Firms reporting quantified outcome metrics (%)	93.5%	62.7%	49.7%	24.5%
Firms naming corporate clients (%)	48.4%	9.2%	5.8%	1.9%
Maximum operational stage: Scaled (%)	94.4%	64.1%	45.5%	17.0%
Maximum operational stage: Deployed (%)	5.6%	31.7%	39.3%	20.8%
Maximum operational stage: Pilot / Exploration (%)	0.0%	1.4%	5.2%	11.3%
Maximum operational stage: No activity reported (%)	0.0%	0.7%	6.3%	49.1%

The full prefilter specification (39 signals across lexical, semantic-anchor, structural, and sentence-level families), the hand-written lexical term lists (18 strong and 12 weak AI terms, roughly 60 named entities), the semantic anchor set (17 positive and 10 negative embedding anchors), feature-importance rankings for the deployed gradient-boosted model, the exact extraction prompts used for semantic-frame classification (prompt v1) and disclosed-activity extraction (prompt v2), throughput and cost figures for the LLM extraction stages, and the deduplication procedure by content hash (8.35 million scorable instances collapsing to 4.87 million unique texts) are documented in the pipeline reference (docs/analytics/10_pipeline.md) and reproduced here for the final frozen configuration (prefilter v2, threshold 0.17, run 20260906T160624Z).

Table 23: Complete granular inference funnel by document form.

Stage	Operational Filter	Unit	10-K	10-Q	DEF 14A	8-K	Calls	Total
A. Corpus								
Documents	Universe coverage	Documents	2,873	8,099	2,646	34,416	7,947	55,981
Paragraphs	Segmentation / turns	Instances	1,544,496	1,736,542	3,679,553	1,623,906	554,281	9,138,778
Scorable paragraphs	Rule: > 3 char & alnum	Instances	1,422,746	1,593,351	3,254,803	1,529,705	553,942	8,354,547
Unique scorable texts	BLAKE2b content hash	Unique texts	855,597	794,548	2,263,881	402,277	483,185	4,799,488
B. Prefilter								
Positive unique texts	GBDT ($\tau = 0.17$, 39 signals)	Unique texts	11,294	2,679	6,344	380	9,584	30,281
Share of scorable texts	Positive rate	%	1.3%	0.3%	0.3%	0.1%	2.0%	0.6%
Positive instances	Exact hash join	Instances	12,487	3,928	6,536	405	9,593	32,952
C. Semantic Frames								
Classified unique texts	Candidate set	Unique texts	11,294	2,679	6,344	380	9,584	30,281
Texts with ≥ 1 frame	LLM judge (Qwen 3.7)	Unique texts	10,532	2,587	5,216	232	7,902	26,469
LLM acceptance rate	Acceptance share	%	93.3%	96.6%	82.2%	61.1%	82.5%	87.4%

Stage	Operational Filter	Unit	10-K	10-Q	DEF 14A	8-K	Calls	Total
Unique semantic frames	Frame extraction	Frames	16,245	3,409	7,168	295	16,249	43,366
Document instances	Exact hash join	Instances	17,758	4,590	7,284	313	16,270	46,215
Documents with ≥ 1 frame	Document join	Documents	1,681	1,431	1,090	206	2,709	7,117
Share of form documents	Document share	%	58.5%	17.7%	41.2%	0.6%	34.1%	12.7%
D. Disclosed Activities								
Texts with firm conduct	Rule: subject = firm	Unique texts	5,927	1,829	2,923	134	6,803	17,420
Share of framed texts	Behaviour share	%	56.3%	70.7%	56.0%	57.8%	86.1%	65.8%
Unique activities	LLM pass 2 (firm ID)	Activities	8,556	2,408	2,670	163	14,020	27,566
Activities per text	Activity density	Ratio	1.44	1.32	0.91	1.22	2.06	1.58
Document instances	Exact hash join	Instances	9,224	3,238	2,708	165	14,033	29,368
Documents with ≥ 1 activity	Document join	Documents	1,298	961	586	88	2,444	5,377
Firms with ≥ 1 activity	Ticker join	Firms	409	159	269	55	383	471 (of 510)

Notes: Complete granular funnel executed on duckdb/thesis.duckdb. Unique texts across columns do not sum to total because identical text paragraphs appear across multiple document forms.

The full distribution of disclosed activities across all six dimensions (27,566 activities, [Table 24](#) below) underlies the activity-evolution chart in Chapter 3 (“Aggregate behavioral landscape”).

Table 24: Full distribution of disclosed activities across all six dimensions.

Action	%	Maturity stage	%	Target audience	%
deploy	62.7%	deployed	58.2%	customers	48.5%
invest infrastructure	7.9%	scaled	14.0%	internal process	25.5%
scale	7.4%	piloting	10.3%	unspecified	13.7%
pilot or explore	4.0%	exploring	9.2%	employees	8.6%
integrate	3.5%	unspecified	8.2%	partners or suppliers	2.5%
develop	3.5%			developers	1.3%
measure outcome	3.4%				
partner	3.0%				
hire or train	2.1%				
buy or license	1.3%				
govern or control	1.2%				
restrict	0.0%				

AI source	%	Evidence strength	%	Business function	%
own	73.0%	generic	53.4%	unspecified	31.9%
unspecified	15.6%	named product or process	29.3%	operations and supply chain	16.5%
third party	7.9%	metric	12.9%	product feature	7.6%
acquired	1.5%	vendor	4.4%	it and cybersecurity	7.0%
mixed	1.0%			other	6.9%
co developed	0.9%			customer service	4.6%
open source	0.1%			software development	4.6%
				marketing and sales	4.1%
				content and media	2.8%
				fraud and risk	2.6%
				data and analytics	2.4%
				research and product development	2.2%
				financial services operations	1.8%
				hr and talent	1.8%
				healthcare delivery	1.1%
				ai compute and models	1.0%
				finance legal and admin	0.8%
				governance	0.5%

Notes: Source: data/processed/clusters/activity_profiles.json. “Business function” has no single dominant category comparable to customer_service for customer-success work specifically. Customer success (retention/expansion) is not coded as a separate family from customer support in the current taxonomy.

APPENDIX B: VALIDATION

The human coding protocol for the pending validation round (300 frame-labeled paragraphs / 434 frames, 300 prefilter-stratified paragraphs, 120 activity-labeled paragraphs / 267 activities, coded in the ui-validator/ tool), per-field agreement statistics once collected, confusion matrices for the promotional and temporal fields, an error taxonomy distinguishing prefilter false positives from judge misclassification, and a sensitivity analysis for collapsing or dropping fields with weak human agreement are reported here as they become available; at the time of writing, only LLM-to-LLM agreement is measured (κ 0.81–0.87, docs/analytics/00_funnel_del_corpus.md).

APPENDIX C: STATISTICAL ROBUSTNESS

Table 25: Full M0-M3 model fit, all five outcomes.

Outcome	R2 M0	R2 M1	R2 M2	R2 M3	Adj.R2 M0	Adj.R2 M1	Adj.R2 M2	Adj.R2 M3
Beta	0.039	0.095	0.125	0.1	-0.098	-0.035	-0.008	-0.033
Idiosyncratic volatility	0.182	0.211	0.225	0.213	0.065	0.098	0.107	0.098
Price-to-sales	0.233	0.235	0.25	0.236	0.125	0.126	0.136	0.124
R&D/sales	0.453	0.479	0.51	0.496	0.409	0.435	0.462	0.451
Revenue growth (t+1)	0.021	0.067	0.116	0.07	-0.122	-0.071	-0.023	-0.071

Notes: Beta and growth start with negative adjusted R^2 at M0 because sector-by-year fixed effects consume many degrees of freedom relative to firms per cell, a penalty the adjustment applies even though raw R^2 stays positive throughout.

Table 26: Standardized coefficients, M2 content block (bold = $p < 0.05$).

Content dimension	Beta	Idiosyncratic volatility	Price-to-sales	R&D/sales	Revenue growth (t+1)
Volume	+0.89 (p=0.02)	+0.51 (p=0.19)	+0.25 (p=0.40)	+0.72 (p<0.01)	-0.07 (p=0.75)
Realized	-0.08 (p=0.82)	-0.31 (p=0.39)	-0.20 (p=0.46)	+0.24 (p=0.42)	+0.09 (p=0.70)
Deployment	-0.55 (p<0.01)	-0.04 (p=0.79)	-0.09 (p=0.57)	-0.37 (p=0.03)	-0.14 (p=0.34)
Capability	-0.13 (p<0.01)	-0.07 (p=0.10)	-0.04 (p=0.55)	-0.11 (p=0.04)	-0.24 (p<0.01)
Risk	-0.20 (p=0.04)	-0.13 (p=0.11)	+0.01 (p=0.92)	-0.26 (p<0.01)	+0.06 (p=0.48)
Governance	-0.13 (p=0.03)	-0.05 (p=0.29)	-0.11 (p=0.04)	-0.11 (p=0.04)	-0.03 (p=0.39)
Promotional	+0.07 (p=0.43)	-0.02 (p=0.81)	-0.09 (p=0.20)	+0.10 (p=0.13)	-0.16 (p=0.02)
Specificity	+0.21 (p=0.11)	+0.24 (p=0.04)	+0.25 (p=0.13)	-0.17 (p=0.12)	+0.64 (p<0.01)

Superseded design, kept for comparison. Chapter 4 originally clustered firms directly on six raw disclosure rates plus a percentile intensity feature; three of them (deployment, capability, adoption stage) declared-action concepts rather than communication style, mixing content the current design deliberately keeps separate for the activity cross-tabulation. The current Chapter 4 clusters on six conceptually named posture dimensions plus intensity, none of them a declared action; stability results for the current design are in Table “Choosing k by bootstrap stability” in Chapter 4. The superseded six-feature run is kept here rather than deleted, per this project’s data-retention policy for anything that required recomputation to produce.

Table 27: Segmentation stability by k, superseded six-feature K-means design.

k	Per-segment Jaccard stability	Minimum	Usable (>=0.60)?
2	0.89, 0.81	0.81	Yes
3	0.70, 0.77, 0.74	0.7	Yes
4	0.57, 0.44, 0.67, 0.70	0.44	No

k	Per-segment Jaccard stability	Minimum	Usable (≥ 0.60)?
5	0.68, 0.39, 0.66, 0.51, 0.35	0.35	No

Robustness to disclosure intensity (superseded design). Re-estimating that six-feature, three-segment solution without disclosure intensity in the clustering features yields an Adjusted Rand Index of 0.910 against the original segmentation; 91.9% of the old “Product Deployers” segment remain in the corresponding cluster, which still shows a substantially higher median disclosure intensity purely as a byproduct of its behavioural profile: high deployment, high customer-facing product share.

The current design does not need this check in the same form, because it separates the concern differently: disclosure intensity is one of the current seven posture dimensions by design (Chapter 4 states why), while every declared-action concept the old design mixed into its clustering (deployment, capability, adoption stage) has been moved out entirely and is used only to characterize the resulting archetypes afterward (with the shared-source-text caveat stated explicitly there), never to build them.

AI-washing index, full specification. The primary estimate is a firm-year weighted least-squares regression, weights = frame count, firm-clustered SE, on 433 firm-years with at least 10 frames (2021–2025): $\text{promo_rate} \sim \text{behavior_index} + \text{grounding_index} + \text{log_volume} + \text{share_def14a} + \text{share_8k} + \text{Sector FE} + \text{Year FE}$. Coefficients: $\text{behavior_index} +0.056$ ($p = 0.064$), $\text{grounding_index} +0.117$ ($p = 0.308$), $\text{log_volume} +0.048$ ($p < 0.001$), $\text{share_def14a} +0.073$ ($p = 0.002$), $\text{share_8k} +0.019$ ($p = 0.79$); $R^2 = 0.377$. A parallel cross-sectional version (pooled 2021–2025, no year FE, $n = 306$ firms) is used for the descriptive comparisons in Chapter 4 and below: $\text{behavior_index} +0.043$ ($p = 0.058$), $\text{grounding_index} +0.198$ ($p = 0.042$), $\text{log_volume} +0.042$ ($p < 0.001$), $\text{share_def14a} +0.068$ ($p = 0.002$), $\text{share_8k} +0.003$ ($p = 0.96$); $R^2 = 0.539$.

This replaces an earlier frame-level logistic design (Poisson-binomial null, intra-document ICC correction, sector-optional controls) that pooled every firm to a single number across all five years and, in its own robustness check, lost its entire flagged tail when dispersion was estimated at the firm level instead of the document level — evidence that what it measured was largely firm heterogeneity, not a construct that moves with anything. The firm-year design fixes this directly: it has real within-firm variation to test (persistence below), and sector and year fixed effects absorb exactly the composition effects the old robustness check was inadvertently detecting.

A placebo test (label permuted within form type, 5 runs) flags on average 0.2 firms as washing under the null, against 13 observed — well-calibrated. A split-half test (frames randomly split per firm, pooled score re-estimated on each half) gives Spearman(z) = 0.403 ($p < 0.001$, $n = 167$ firms with ≥ 10 frames in both halves), and the firm-year panel’s year-over-year persistence (z in year t vs. $t+1$) is Spearman = +0.379 ($p < 0.001$, $n = 198$ pairs): positive and clearly short of 1, consistent with a signal that moves with something real but is not a frozen firm trait.

Table 28: Washing-index sensitivity across three specifications (pooled cross-section, 2021-2025).

Specification	Washing tail (n)	Quiet tail (n)	Jaccard vs. reference
Reference (unique frames, sector FE)	13	4	1
No sector FE	16	4	0.61
Document instances (not deduplicated)	13	6	0.73

Notes: CDNS, CMI, CRWD, HUM, JCI, MS, NOC, PANW, and QCOM are the only firms flagged in the washing tail under all three specifications. Dropping sector fixed effects widens the tail from 13 to 16

firms (Jaccard 0.61), confirming that part of the excess a less-controlled specification would flag is sector composition, not discourse.

Table 29: All firm-years flagged by the AI-washing index (panel specification), with grounding and archetype.

Ticker	Year	Observed promotional frames	Expected (model)	z	Activity grounding	Disclosure archetype (Chapter 4)
LDOS	2025	4	0.4	5.76	0.24	Vocal Substantives
CDNS	2025	18	6.9	4.51	0.41	Vocal Substantives
PANW	2023	18	7.2	4.46	0.47	Vocal Substantives
CRM	2021	5	1	4.14	0.29	Vocal Substantives
CHRW	2025	5	1.1	3.91	0.35	Vocal Substantives
CRM	2022	7	1.9	3.9	0.32	Vocal Substantives
PANW	2024	23	11.8	3.65	0.53	Vocal Substantives
JNPR	2024	14	5.9	3.59	0.48	Vocal Substantives
CMI	2025	2	0.3	3.51	0.15	Defensive Disclosers
QCOM	2025	13	5.4	3.43	0.33	Vocal Substantives
QCOM	2024	12	5	3.34	0.28	Vocal Substantives
CRWD	2025	9	3.4	3.21	0.54	Vocal Substantives

Notes: Rows with positive z are the washing tail (promotes more than expected); the panel specification flags no firm-years with negative z at 5% FDR, unlike the pooled cross-sectional version, which flags four quiet-tail firms (META, MSCI, V, MSI). Most washing-tail firm-years sit in the Vocal Substantives archetype (Chapter 4); a firm has to already show substantial disclosed activity to register as promoting *in excess* of that baseline.

Alternative firm-year aggregation. Results are unchanged when firm-year intensity is constructed as an aggregate-numerator-over-aggregate-denominator ratio rather than the median document rate: the two rollups are numerically identical across all seven content dimensions and 2,475 firm-years (max absolute difference below 1e-9), because a firm-year's AI content comes from a single dominant filing often enough that the rollup rule carries no weight in practice.

APPENDIX D: VARIABLE CONSTRUCTION

Table 30: XBRL concept coverage.

Financial concept	Coverage of the firm-year panel
Revenue	98%
Net margin	98%
Capex	87%
SG&A	80%
R&D expense	45% (excluded from fundamentals; used only as an outcome)

Fiscal-period alignment rules (period-end date matched to the nearest prior filing date; the [340, 380]-day validity window for year-over-year growth), the market-model beta and idiosyncratic-volatility construction (252 trading days pre-filing, one-factor market model), and the ROIC-minus-WACC construction (geometric equity risk premium of 6.48%, not arithmetic) are documented here.

APPENDIX E: SEC / DEEPSEEK SECONDARY SPECIFICATIONS

Secondary SEC event-study specifications (alternative window widths, weighted variants, cells excluding mixed pre/post filings) and alternative event-window specifications for both shocks are

reported here in full, consistent with the one-sentence summary in Chapter 6: no discontinuity separable from prior trends was identified for DeepSeek, and the analysis is not extended to outcome variables (competitive response, generative-AI capex) that a valid identification would be needed to interpret.

Table 31: DeepSeek event study (2025Q1 cutoff, 5,287 firm-quarters, 484 firms): pre-trend test by outcome.

Outcome	Pre-trend test p-value	Fails at 5%?
Promotional	0.017	Yes
Specificity	<0.001	Yes
Risk	<0.001	Yes
Hypothetical framing	0.009	Yes
Total AI frames	<0.001	Yes

Notes: All five outcomes fail the pre-trend test, so no discontinuity around the January 2025 shock can be separated from a pre-existing trend for any of them.